DBEDML: Dual-Branch Energy-Driven Meta-Learning for Hyperspectral Imagery Cross-Domain Few-Shot Open-Set Recognition

Qixing Yu, Zhongwei Li, Fangming Guo, *Member, IEEE*, Ziqi Xin, Guangbo Ren, Jianbu Wang, and Zhenggang Bi

***Abstract*—In the true hyperspectral scenes, open-set recognition (OSR) is inevitable, yet large-scale label acquisition is challenging due to environmental complexity. An intriguing approach is to transfer knowledge from scenes with large-scale labeled samples (source domain, SD) to target scenes (target domain, TD), then fine-tune using a few labeled TD samples to recognize known classes and reject unknown ones. However, three issues arise, namely insufficient class samples, asymmetrical interdomain label spaces, and the open-set learning problem. To address the aforementioned problems, we define this as the cross-domain few-shot OSR (CDFSOSR) task and propose a dual-branch energy-driven meta-learning (DBEDML) algorithm. For problems 1 and 3, we design a dual-branch architecture incorporating contrastive learning and distance metrics, driven by an energy function, to effectively solve OSR in few-shot learning (FSL). For problem 2, we introduce fake prototypes as unknown classes in SD for domain alignment (DA), and design a decoupled adversarial DA module to align known class distributions between SD and TD while distinguishing unknown classes. Extensive experiments on three self-constructed coastal wetland datasets and two public datasets demonstrate the effectiveness of DBEDML. Code will be available at https://github.com/Yqx-ACE/TGRS_2026_DBEDML**

***Index Terms*—Cross-domain, few-shot learning (FSL), hyperspectral image (HSI) classification, meta-learning, open-set recognition (OSR).**

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Qixing Yu, Zhongwei Li, Fangming Guo, Ziqi Xin, and Jianbu Wang are with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao 266580, China (e-mail: yuqixing.upc@vip.163.com; li.zhongwei@vip.163.com; guofangming@upc.edu.cn; B21160015@s.upc.edu.cn; wangjianbu@fio.org.cn).

Guangbo Ren is with the Laboratory of Marine Physics and Remote Sensing, First Institute of Oceanography, Ministry of Natural Resources, Qingdao 266061, China, and also with Qingdao Key Laboratory of Spatial Information Dynamic Perception and Smart Cartography in the Yellow River Basin (Shandong), Qingdao 266000, China (e-mail: renguangbo@126.com).

Zhenggang Bi is with the Yellow River Estuary Management Station, Shandong Yellow River Delta National Nature Reserve, Dongying 257500, China (e-mail: hhkbzg@163.com).

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I. Introduction

HYPERSPECTRAL images (HSIs) have emerged as a key tool for coastal wetland vegetation observation due to their spectral fingerprint capabilities, capturing diagnostic absorption/reflection characteristics from visible to shortwave infrared bands [1], [2], [3]. However, environmental complexity and high spatial resolution make manual labeling expertise-dependent and costly, limiting supervised HSI classification (HSIC) methods that require extensive labeled datasets [4], [5].

To reduce reliance on labels, semi-supervised and unsupervised HSIC algorithms have been developed. Zhan et al. [6] introduced a spectral angle-based generative adversarial network with multilayer feature fusion to guide discriminator learning from limited labels. Han et al. [7] constructed a multiscale adjacency graph leveraging relationships between labeled and unlabeled samples to enhance classification accuracy. While these methods expand data utilization, they still require a certain number of high-quality anchor labels, remain sensitive to label noise, and cannot ensure stable performance.

Unsupervised domain adaptation (UDA) requires no target domain (TD) sample labels during training, aligning distributions using labeled source domain (SD) data and unlabeled TD data [8]. This eliminates the need for TD labels and reduces performance loss from cross-domain discrepancies, providing a practical solution to label scarcity in HSIC. As shown in Fig. 1(a), ideal UDA assumes SD and TD share the same closed-set label space with abundant samples. The goal is to align feature distributions across domains by minimizing distribution divergence for accurate classification. For example, Huang et al. [9] developed a spatial–spectral weighted adversarial domain adaptation network for wetland mapping with HSIs, while Li et al. [10] addressed spectral shift via a cyclic self-training framework to improve cross-scene classification in coastal wetlands. However, these ideal conditions are rarely met in practice. It mainly faces the following three problems.

1) *Insufficient Class Samples Problem:* The complexity of coastal wetlands makes obtaining sufficient labeled class samples difficult, limiting the extraction of domain knowledge across different areas [11]. In addition,

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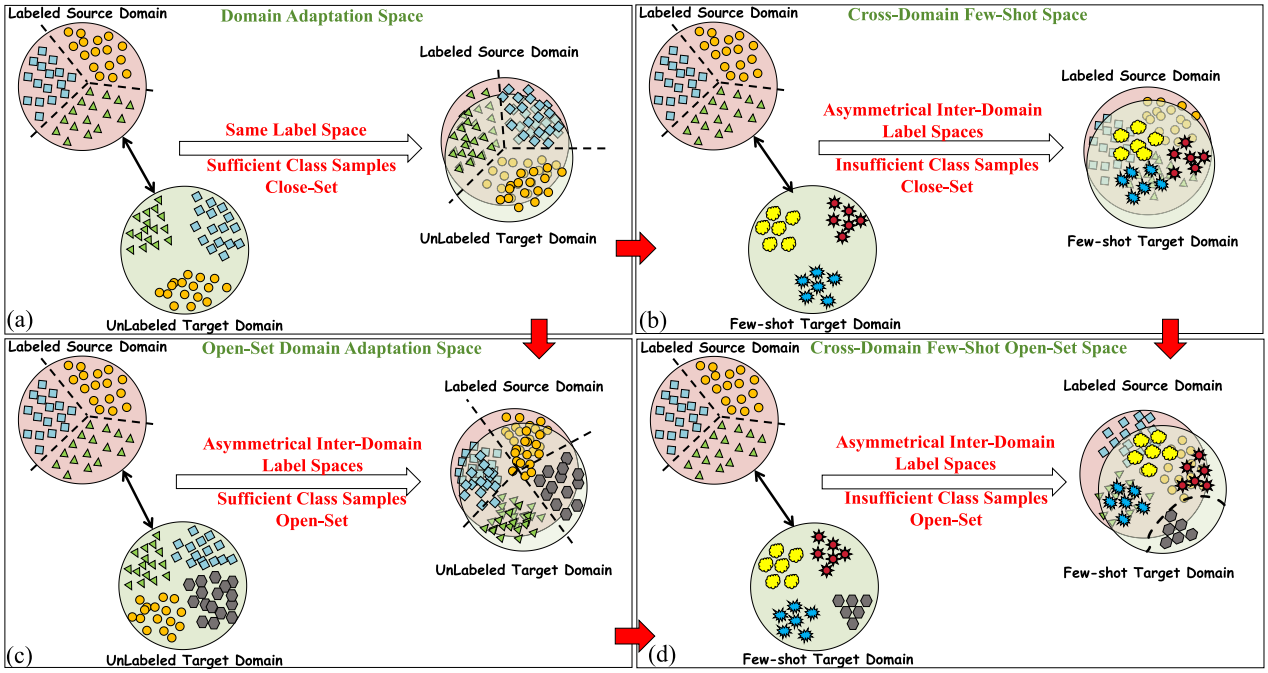


Fig. 1. Comparison diagrams of four task spaces. (a) TD consists solely of unlabeled samples from the same categories as the SD. (b) TD has entirely nonoverlapping classes with the SD, and only a few labeled samples are provided per class in TD. (c) TD includes unlabeled samples from both known classes (shared with the SD) and unknown classes (unique to the TD). (d) TD includes both known and unknown classes with entirely disjoint label spaces from the SD. Only a few labeled samples are provided per known class in TD.

acquiring these samples demands extensive professional expertise.

2) *Asymmetrical Interdomain Label Spaces Problem:* Many UDA methods assume that the SD and TD share consistent label spaces [12]. However, in practice, acquiring sufficient labels for the SD is challenging, so researchers often use other well-labeled domains as the SD [13]. This can cause partial or complete label space inconsistency between the SD and TD. Blindly aligning feature distributions may cause negative transfer and misclassification. This issue also appears in open-set learning, discussed below.
3) *Open-Set Learning Problem:* In coastal wetland HSI scenes, it is challenging to cover all classes during training. When classifying all HSI pixels, unknown-class samples are often misclassified as known classes, misleading vegetation monitoring and ecological protection. Thus, the algorithm must identify unknown classes as "Unknown" during testing to distinguish them from known classes [14].

Building on UDA, several tasks have been proposed to address its challenges. When UDA faces both problems 1 and 2, it is termed cross-domain few-shot learning (CDFSL), as shown in Fig. 1(b). CDFSL typically alleviates problem 1 via meta-knowledge transfer and mitigates problem 2 by minimizing domain distribution divergence, aiming for overall distribution alignment. Numerous studies have explored this area. For instance, Li et al. [15] first addressed CDFSL with the DCFSL algorithm, establishing a foundational framework. Later, Liu et al. [16] proposed ADLKAN, which enhances sparse feature capture using a multiscale conditional domain discriminator to strengthen few-shot learning (FSL) under domain shift. Notably, the CDFSL task does not involve open-set learning.

When UDA faces problems 2 and 3, it becomes open-set domain adaptation (OSDA), as shown in Fig. 1(c). OSDA aims to minimize domain distribution divergence (problem 2) and distinguish unknown classes (problem 3). The goal is to reject unknown classes while aligning known classes' feature distributions. Bi et al. [17] proposed a weighted generative adversarial network with a dynamic threshold to learn the classification boundary between known and unknown classes in feature space. Wang et al. [18] introduced a teacher–student critical learning module with a distillation mechanism to address interclass variability, which threshold-based methods often miss. Although the TD is unlabeled, it shares the label space with the SD except for unknown classes and has sufficient samples per class, so problem 1 is not involved.

In coastal wetland scenes, HSIs often require vegetation recognition with only a few labeled samples per class. All classes except the target ones must be identified as "Unknown." Researchers seek to transfer knowledge from a well-labeled SD HSI to a TD HSI, achieving accurate classification despite unknown classes and limited TD labels. This article defines this challenge as the cross-domain few-shot open-set recognition (CDFSOSR) task. Existing UDA, CDFSL, and OSDA algorithms struggle with CDFSOSR, highlighting the need for an HSIC algorithm that addresses all three issues simultaneously. It is worth noting that previous works such as OCN [19] and HyperTaFOR [20] have conducted research on the few-shot open-set recognition (FSOSR) task, but none of these works have considered the impact of domain distribution differences, which is also the difference between CDFSOSR and FSOSR.

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To address these issues, we propose dual-branch energy-driven meta-learning (DBEDML), an algorithm for the CDFSOSR task based on a meta-learning framework with meta-training and meta-testing phases, where the SD and TD alternately undergo meta-training. To simulate an open-set environment, each domain serves as an open set for the other. During FSL, a dual-branch energy-driven architecture generates fake prototypes from embedding and feature distributions perspectives. Through contrastive learning and distance metrics, it minimizes distances between real prototypes and known classes, reduces gaps between fake prototypes and unknown classes, and maximizes separation between real and fake prototypes. The energy function integrates metrics from both branches, assigning higher energy scores to unknown samples for accurate recognition. For domain alignment (DA) under open-set conditions, since SD and TD label spaces largely differ, directly aligning known classes' feature distributions while rejecting unknown classes of TD is challenging. We treat fake prototypes generated in the SD as unknown-class representations and involve them in DA. A decoupling strategy separates both SD and TD features into known and unknown categories, which are aligned separately via adversarial learning, achieving DA while excluding unknown target classes. Building on the success of Mamba [21] in HSIC, we adopt it as the core feature extractor and propose a novel slice-and-scan strategy that alternates scanning within and across slices, enhancing spatial dependencies without significant computational overhead. The main contributions of DBEDML are as follows.

1) We focus on the CDFSOSR task by proposing the DBEDML architecture, which employs a meta-learning strategy. Extensive experiments on five datasets demonstrate the superiority of DBEDML.
2) We design a dual-branch energy-driven model to address the open-set problem in the FSL process. By employing contrastive learning and distance metric strategies, the model integrates real prototypes with known class features while disentangling fake prototypes. In addition, an energy function is designed to drive the metric distances of the two branches, rendering unknown samples more distinguishable.
3) We utilize the fake feature prototypes from the SD as unknown classes to aid the DA process, mitigating challenges from label space inconsistencies between the SD and TD. We design a decoupling strategy and employ an adversarial approach to separately align feature distributions of known and unknown classes in both domains, effectively aligning known domain distributions while excluding unknown classes.
4) We propose a Mamba-based feature extractor and a novel slice scanning strategy to address spatial misalignment caused by conventional methods, achieving effective feature extraction with reasonable time cost.

The remaining sections are as follows. Section II reviews the relevant literature. Section III details the design specifics of DBEDML. Section IV analyzes the experimental findings and carries out a set of ablation studies to verify the effectiveness of each component in DBEDML. Section V offers concluding remarks and future perspectives.

II. Related Work

A. FSOSR Task

FSOSR, a precursor to CDFSOSR, integrates open-set recognition (OSR) into the FSL framework, enabling models to identify unknown classes during FSL. Common OSR approaches include score-based, distance-based, and reconstruction-based approaches [22], [23]. Score-based methods, such as Bendale and Boult [24], use extreme value theory to model distances between activation vectors and class means, albeit with high computational costs. Vaze et al. [25] improved results by leveraging translation invariance in logit vectors. Distance-based methods emphasize feature compactness within known classes and discrimination between them. Yang et al. [26] proposed convolutional prototype learning with prototype loss to cluster features around class prototypes, while Miller et al. [27] clustered logit vectors around class anchors. Reconstruction-based methods assume known classes have lower reconstruction error. Yoshihashi et al. [28] developed an autoencoder trained with classification and reconstruction losses, using classification scores to separate known and unknown samples. Since OSR typically requires ample labeled data, adapting it to data-scarce FSL settings has motivated ongoing research.

In FSOSR, Zhang et al. [29] developed a framework for automatic modulation classification using meta-prototype training for feature extraction and a modular open-set classifier to identify known and unknown modulation types. Pal et al. [19] focused on HSIs, jointly optimizing an outlier calibration network and feature extractor during meta-training, achieving strong results. Xi et al. [20] proposed HyperTaFOR, which integrates a spatial–spectral selective Transformer to achieve task-adaptive FSOSR. However, these FSOSR methods overlook feature distribution shifts caused by domain divergence, which significantly degrades classification accuracy and motivates our work on CDFSOSR.

B. OSDA Task

The OSDA task has evolved by integrating OSR into domain adaptation, aiming to transfer knowledge from a well-labeled SD to an unlabeled TD under a shared label space, while rejecting unknown classes. In OSDA, Tang et al. [30] employed adaptive discriminative graph learning to capture the intrinsic manifold structure of both SD and TD in a latent feature space, thereby clarifying decision boundaries and mapping samples into a high-dimensional space to achieve linear separability, though this introduces a nonconvex optimization problem. To mitigate the common issue of negative transfer in OSDA, Tian et al. [31] incorporated optimal transport and adversarial learning to introduce a similarity matrix for feature-to-prototype clustering, enhancing the ability to capture discriminative information. Chang et al. [32] proposed a mutual-to-separate framework to handle larger domain gaps, which consists of a sample separation network and a distribution matching network that mutually supervise and alternate to improve alignment in the shared label space.

However, OSDA assumes that the SD and TD largely share the same label space, with the SD simply lacking

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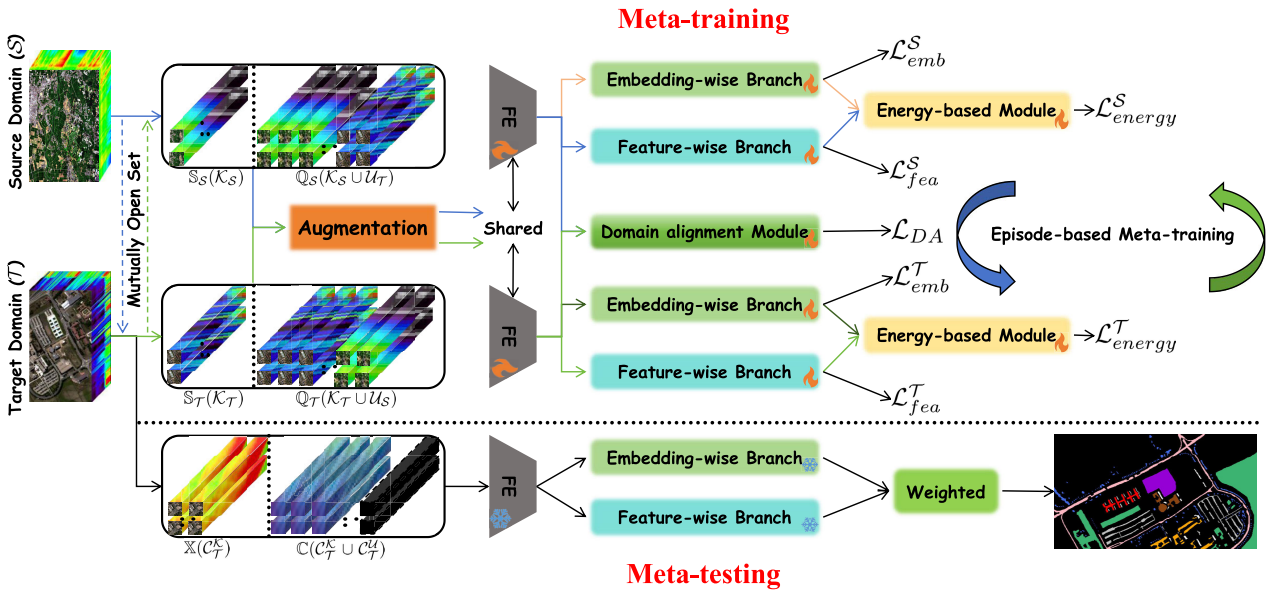


Fig. 2. Proposed DBEDML framework comprises meta-training and meta-testing phases. During meta-training, SD and TD data are organized into support and query sets, with each domain acting as an open set for the other. After augmentation, features are extracted and passed through a dual-branch energy-driven network, generating losses $\mathcal{L}_{emb}^{S/T}$, $\mathcal{L}_{fea}^{S/T}$, and $\mathcal{L}_{energy}^{S/T}$. Features then enter the DA module, producing $\mathcal{L}_{DA}$. In the meta-testing phase, TD data are split into a training set and a test set containing unknown classes. Processed by a frozen feature extractor, the data are input to the dual-branch architecture, and the final classification is obtained by weighting the output results.

some unknown classes present in the TD. This allows class labels from the SD to still be utilized during adaptation. In contrast, under the CDFSOSR setting, the SD and TD have disjoint label spaces, and there are no unknown classes in the SD. Therefore, a critical challenge remains: how to align the feature distributions of known classes from the TD with those of the SD, while simultaneously pushing away the distribution of unknown classes from the TD during domain adaptation.

III. Methodology

The overall architecture of the proposed DBEDML, which consists of the meta-training and meta-testing phases, is illustrated in Fig. 2. The detailed elaboration is presented in Sections III-A–III-E.

A. Problem Definition and Notations

In the CDFSOSR task, the set of classes in the SD is denoted as $\mathcal{C}_{\mathcal{S}}$, and that in the TD as $\mathcal{C}_{\mathcal{T}} = \mathcal{C}_{\mathcal{T}}^{\mathcal{K}} \cup \mathcal{C}_{\mathcal{T}}^{\mathcal{U}}$, with the constraint that $|\mathcal{C}_{\mathcal{S}}| \geq |\mathcal{C}_{\mathcal{T}}^{\mathcal{K}}|$, where $|\mathcal{C}_{\mathcal{S}}|$ represents the length of $\mathcal{C}_{\mathcal{S}}$. An episodic training strategy is employed, where each episode samples N examples from each of the $|\mathcal{C}_{\mathcal{T}}^{\mathcal{K}}|$ classes, that is $|\mathcal{C}_{\mathcal{T}}^{\mathcal{K}}|$-Way N-shot. In the proposed setup, taking the SD as an example, each episode consists of a set of known classes $\mathcal{K}_{\mathcal{S}}$ from the SD and a set of pseudo-unknown classes $\mathcal{U}_{\mathcal{T}}$ from the TD, where $\mathcal{K}_{\mathcal{S}} \subset \mathcal{C}_{\mathcal{S}}, \mathcal{U}_{\mathcal{T}} = \mathcal{C}_{\mathcal{T}}^{\mathcal{K}}$, and $|\mathcal{K}_{\mathcal{S}}| = |\mathcal{C}_{\mathcal{T}}^{\mathcal{K}}|$. Similarly, the known classes and pseudo-unknown classes in the TD are denoted as $\mathcal{K}_{\mathcal{T}}$ and $\mathcal{U}_{\mathcal{S}}$, respectively, with $\mathcal{K}_{\mathcal{T}} = \mathcal{C}_{\mathcal{T}}^{\mathcal{K}}$, $\mathcal{U}_{\mathcal{S}} \subset \mathcal{C}_{\mathcal{S}}$, and $|\mathcal{U}_{\mathcal{S}}| = |\mathcal{C}_{\mathcal{T}}^{\mathcal{K}}|$. Furthermore, $\mathcal{K}_{\mathcal{S}} \cap \mathcal{U}_{\mathcal{T}} = \varnothing$, $\mathcal{K}_{\mathcal{T}} \cap \mathcal{U}_{\mathcal{S}} = \varnothing$, while the relationship between $\mathcal{K}_{\mathcal{S}}$ and $\mathcal{U}_{\mathcal{S}}$ is not predetermined.

To construct the support set $\mathbb{S}_{\mathcal{S}}$ and the query set $\mathbb{Q}_{\mathcal{S}}$, taking the SD as an example, $m_{\mathcal{S}}$ samples are selected from each class in $\mathcal{K}_{\mathcal{S}}$, of which $m_{\mathcal{S}}^s$ samples are allocated to $\mathbb{S}_{\mathcal{S}}$ and $m_{\mathcal{S}}^q$ samples to $\mathbb{Q}_{\mathcal{S}}$, such that $m_{\mathcal{S}}^s + m_{\mathcal{S}}^q = m_{\mathcal{S}}$. In addition to the $m_{\mathcal{S}}^q$ samples from the SD, the $\mathbb{Q}_{\mathcal{S}}$ also includes $m_{\mathcal{S}}$ samples drawn from the $\mathcal{U}_{\mathcal{T}}$ pseudo-unknown classes. In summary, the $\mathbb{S}_{\mathcal{S}}$ can be formally expressed as $\mathbb{S}_{\mathcal{S}}(\mathcal{K}_{\mathcal{S}}) = \{x_i^s, y_i^s\}_{i=1}^{|\mathcal{K}_{\mathcal{S}}|*m_{\mathcal{S}}^s}$, and the $\mathbb{Q}_{\mathcal{S}}$ is that $\mathbb{Q}_{\mathcal{S}}(\mathcal{K}_{\mathcal{S}} \cup \mathcal{U}_{\mathcal{T}}) = \{x_i^s, y_i^s\}_{i=1}^{|\mathcal{K}_{\mathcal{S}}|*m_{\mathcal{S}}^q} \cup \{x_i^t\}_{i=1}^{|\mathcal{U}_{\mathcal{T}}|*m_{\mathcal{S}}}$. Similarity, $\mathbb{S}_{\mathcal{T}}(\mathcal{K}_{\mathcal{T}}) = \{x_i^t, y_i^t\}_{i=1}^{|\mathcal{K}_{\mathcal{T}}|*m_{\mathcal{T}}^s}$ and $\mathbb{Q}_{\mathcal{T}}(\mathcal{K}_{\mathcal{T}} \cup \mathcal{U}_{\mathcal{S}}) = \{x_i^t, y_i^t\}_{i=1}^{|\mathcal{K}_{\mathcal{T}}|*m_{\mathcal{T}}^q} \cup \{x_i^s\}_{i=1}^{|\mathcal{U}_{\mathcal{S}}|*m_{\mathcal{T}}}$. The SD and TD employ a symmetric configuration with $m_{\mathcal{S}} = m_{\mathcal{T}}$, $m_{\mathcal{S}}^s = m_{\mathcal{T}}^s$ and $m_{\mathcal{S}}^q = m_{\mathcal{T}}^q$.

During the meta-testing phase, a subset of samples from $\mathcal{C}_{\mathcal{T}}^{\mathcal{K}}$ is partitioned into the training set $\mathbb{X}(\mathcal{C}_{\mathcal{T}}^{\mathcal{K}})$ to simulate the support set used in the meta-training stage, while the remaining samples from $\mathcal{C}_{\mathcal{T}}^{\mathcal{K}}$ and all samples from $\mathcal{C}_{\mathcal{T}}^{\mathcal{U}}$ are assigned to the test set $\mathbb{C}(\mathcal{C}_{\mathcal{T}}^{\mathcal{K}} \cup \mathcal{C}_{\mathcal{T}}^{\mathcal{U}})$, emulating the query set. The final classification outcome is derived from a weighted combination produced by the dual-branch architecture.

B. HVSE Based on Mamba

The selective state space model (Mamba-S6) has demonstrated strong performance in HSIC. Pan et al. [33] proposed MambaLG to address limitations in receptive field size and quadratic complexity, while Li et al. [34] designed spatial and spectral Mamba blocks to model long-range spatial–spectral dependencies. However, most existing Mamba-based methods still adopt a "flatten-and-scan" strategy, which often fails to balance spatial alignment and computational efficiency. Inspired by MambaIR [35] and XYScanNet [36], we introduce a novel slice-scanning strategy and develop a horizontal–vertical scanning extractor (HVSE) to achieve accurate feature representation for HSIs. The structure of HVSE is illustrated in Fig. 3(a). HVSE is designed to progressively extract spatial–spectral information from input samples, from shallow to deep levels. Its core component is an Encoder Block, whose architecture is shown in Fig. 3(b). The

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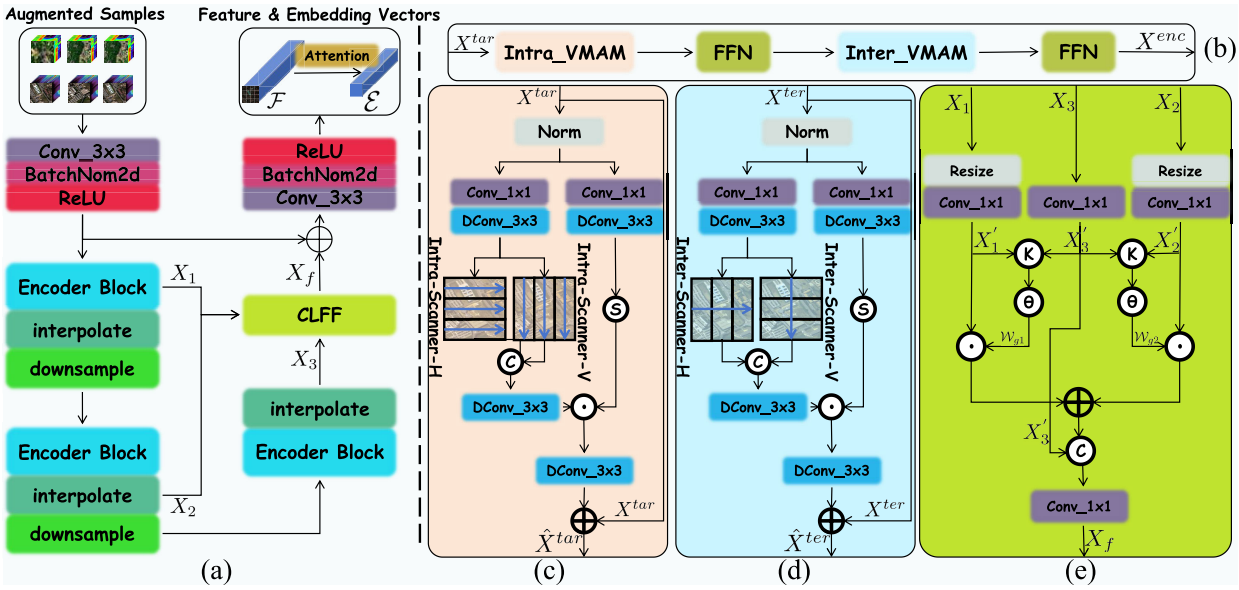


Fig. 3. (a) Overall architecture of the HVSE. (b) Encoder block. (c) Intra-VMAM block. (d) Inter-VMAM block. (e) CLFF block.

block comprises two feed-forward neural networks (FFNs), an intraslice scanning Mamba (Intra-VMAM) module, and an interslice scanning Mamba (Inter-VMAM) module.

1) Intra-VMAM Module: The Intra-VSSM module is designed to establish long-range dependencies between any pair of pixels across spatial dimensions, thereby effectively granting each pixel a global receptive field. Given an input feature map $X^{tar} \in \mathbb{R}^{B\times C\times H\times W}$, it is first normalized and then processed by a 1×1 convolution ($Conv_{1\times 1}$) that doubles the number of channels. A 3×3 depthwise separable convolution ($DConv_{3\times 3}$) is applied, and the output is split along the channel dimension into two parts: one, denoted as $F^{tar} \in \mathbb{R}^{B\times C\times H\times W}$, is used for scanning, while the other, $Y^{tar} \in \mathbb{R}^{B\times C\times H\times W}$, is retained as a gating signal. The scanning feature F^{tar} is further divided into a horizontal branch feature $F_h^{tar} \in \mathbb{R}^{B\times C/2\times H\times W}$ and a vertical branch feature $F_v^{tar} \in \mathbb{R}^{B\times C/2\times H\times W}$. For F_h^{tar}, a row-wise slicing strategy is applied: it is transposed to obtain $F_h^{tar\top}$, and the batch B and height H dimensions are merged, forming $B\times H$ independent sequences each of length W with feature dimension $C/2$. These sequences are processed by the Mamba-S6 module and reshaped to produce $\hat{F}_h^{tar} \in \mathbb{R}^{B\times C/2\times H\times W}$. Similarly, F_v^{tar} undergoes column-wise slicing: the batch B and width W dimensions are merged, resulting in a sequence of shape $[B\times W, H, C/2]$, which is processed by Mamba-S6 and reshaped back into $\hat{F}_v^{tar} \in \mathbb{R}^{B\times C/2\times H\times W}$. The features $\hat{F}_h^{tar}$ and $\hat{F}_v^{tar}$ are then concatenated along the channel dimension and fused via a $DConv_{3\times 3}$, yielding $\hat{F}_{hv}^{tar} \in \mathbb{R}^{B\times C\times H\times W}$. The gating signal Y^{tar} is activated by the SiLU function and applied to $\hat{F}_{hv}^{tar}$ via Hadamard product for modulation. Finally, a $DConv_{3\times 3}$ is used to transform the features, producing the output $\hat{X}^{tar} \in \mathbb{R}^{B\times C\times H\times W}$ with X^{tar}.

2) Inter-VMAM Module: The Inter-VMAM module is designed to perform adaptive weighting along the spectral dimension by analyzing the global importance of each spectral feature, thereby recalibrating the spectral weights to emphasize informative features. Given an input feature map $X^{ter} \in \mathbb{R}^{B\times C\times H\times W}$, the initial operations mirror those of Intra-VMAM, producing a scanning feature $F^{ter} \in \mathbb{R}^{B\times C\times H\times W}$ and a gating signal $Y^{ter} \in \mathbb{R}^{B\times C\times H\times W}$. The tensor F^{ter} is split along the channel dimension into horizontal and vertical scanning features $F_h^{ter} \in \mathbb{R}^{B\times C/2\times H\times W}$ and $F_v^{ter} \in \mathbb{R}^{B\times C/2\times H\times W}$, respectively. For the horizontal branch, a row-wise slicing strategy captures relationships among spectral bands and their importance across rows. Specifically, F_h^{ter} is reshaped to $[B, H, C/2, W]$, followed by global average pooling (GAP) along the width dimension W. The resulting tensor is reshaped to $[B, H, C/2]$, representing B batches of H independent sequences of length $C/2$, where each element corresponds to the global average response of a spectral feature. These sequences are processed by Mamba-S6, and the output is passed through a Sigmoid function to generate weight scores $\mathcal{W}_h \in \mathbb{R}^{B\times H\times C/2}$ in the range [0, 1]. The tensor F_h^{ter} is then weighted by $\mathcal{W}_h$ and adjusted in dimensionality to produce the branch output $\hat{F}_h^{ter} \in \mathbb{R}^{B\times C/2\times H\times W}$. The vertical branch follows a symmetric process along the height dimension H, exploring importance variations across columns, and yields output $\hat{F}_v^{ter} \in \mathbb{R}^{B\times C/2\times H\times W}$. The features $\hat{F}_h^{ter}$ and $\hat{F}_v^{ter}$, along with the gating signal Y^{ter}, are processed as in Intra-VMAM, resulting in the final output $\hat{X}^{ter} \in \mathbb{R}^{B\times C\times H\times W}$ of Inter-VMAM with X^{ter}.

The HVSE module contains three encoder blocks that hierarchically extract multilevel features. A cross-layer feature fusion (CLFF) module adaptively integrates deep semantic information with shallow spatial details, producing a discriminative representation combining fine-grained spatial and high-level semantic cues, as shown in Fig. 3(e). Denote the outputs of the Encoder Blocks from shallow to deep as $X_1 \in \mathbb{R}^{B\times C\times H\times W}$, $X_2 \in \mathbb{R}^{B\times 2C\times H/2\times W/2}$, and $X_3 \in \mathbb{R}^{B\times 4C\times H/4\times W/4}$, respectively. Using X_3 as the reference, the three features are first projected to a common feature space. Then, X_1 and X_2 are upsampled to match the spatial dimensions of X_3. The divergence between X_3 and X_1, as well as between X_3 and X_2, is measured using Kullback–Leibler (KL) divergence. Gating

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weights are then generated via a sigmoid function based on these values. The detailed procedure is described below

$$\mathcal{W}_{g1} = \theta\left(D_{KL}\left(\text{LogSigmoid}\left(X_1\right)\|\text{LogSigmoid}\left(X_3\right)\right)\right)$$
$$\mathcal{W}_{g2} = \theta\left(D_{KL}\left(\text{LogSigmoid}\left(X_2\right)\|\text{LogSigmoid}\left(X_3\right)\right)\right) \quad (1)$$

where θ denotes the sigmoid function, and $D_{KL}(\cdot\|\cdot)$ represents the KL divergence. The gating weights are applied to weight features X_1 and X_2, which are then concatenated with X_3 along the channel dimension. $Conv_{1\times1}$ is utilized to optimally fuse these features, producing the final output $X_f \in \mathbb{R}^{B\times C\times H\times W}$, where each pixel is enhanced by deep semantic information.

The output of CLFF is combined with the original input via a residual connection. A convolutional layer is then applied to adjust the spatial dimensions, yielding the feature vector $\mathcal{F} \in \mathbb{R}^{B\times\hat{C}\times\hat{H}\times\hat{W}}$. An attention pooling operation is applied to $\mathcal{F}$ to obtain the embedded vector $\mathcal{E} \in \mathbb{R}^{B\times\hat{C}}$, as described in the following procedure:

$$\mathcal{E}_{b,\hat{c}} = \sum_{\hat{h}=1}^{\hat{H}}\sum_{\hat{w}=1}^{\hat{W}}\left(\mathcal{F}_{b,\hat{c},\hat{h},\hat{w}}\cdot\theta\left(\sum_{\hat{c}^*=1}^{\hat{C}} w_{\hat{c}^*}\cdot\mathcal{F}_{b,\hat{c}^*,\hat{h},\hat{w}} + bias\right)\right). \quad (2)$$

All feature vectors and embedded vectors are integrated according to the support set or query set they belong to, and are denoted as $\mathbb{S}_{\mathcal{F}} \in \mathbb{R}^{|\mathcal{C}_{\mathcal{T}}^{\mathcal{K}}|\times m^s\times\hat{C}\times\hat{H}\times\hat{W}}$, $\mathbb{S}_{\mathcal{E}} \in \mathbb{R}^{|\mathcal{C}_{\mathcal{T}}^{\mathcal{K}}|\times m^s\times\hat{C}}$, $\mathbb{Q}_{\mathcal{F}} \in \mathbb{R}^{\hat{B}\times\hat{C}\times\hat{H}\times\hat{W}}$, and $\mathbb{Q}_{\mathcal{E}} \in \mathbb{R}^{\hat{B}\times\hat{C}}$, respectively. $\hat{B}$ denotes the number of all samples in the query set.

C. Dual-Branch Energy-Driven Architecture

The dual-branch energy-driven architecture is aimed at accurately identifying known classes and delineating known-unknown class boundaries. It consists of three components. The embedding-wise branch (EWB), which focuses on globally discriminative spectral features, the feature-wise branch (FWB), which captures local contextual information such as texture and shape, and the energy-based module (EBM), which integrates both branches by emphasizing potential unknown-class samples to sharpen interclass boundaries. Due to the consistency between SD and TD, the following description uses SD as an example.

1) Embedding-Wise Branch: As shown in Fig. 4(a), the EWB schematic is divided by a dashed line into two sections. The upper section illustrates the process of generating support embedding prototypes $\mathbb{S}_{\mathcal{E}}^{\mathcal{P}} \in \mathbb{R}^{|\mathcal{K}_{\mathcal{S}}|\times\hat{C}}$ from the $\mathbb{S}_{\mathcal{E}}$, followed by the creation of fake embedding prototype $\mathbb{F}_{\mathcal{E}}^{\mathcal{P}} \in \mathbb{R}^{1\times\hat{C}}$. The lower section represents contrastive learning and distance metric computation between the $\mathbb{Q}_{\mathcal{E}}$ and both the $\mathbb{S}_{\mathcal{E}}^{\mathcal{P}}$ and $\mathbb{F}_{\mathcal{E}}^{\mathcal{P}}$.

In prototype networks, class prototypes are typically generated by averaging features, yet such prototypes often lack discriminative power. Therefore, a learnable calibration network $G_n(\cdot)$ is introduced to refine prototype generation, enhancing interclass discriminability. The procedure is described below

$$\mathbb{S}_{\mathcal{E}}^{\mathcal{P}} = G_n\left(\frac{1}{|\mathcal{K}_{\mathcal{S}}|}\sum_{k=1}^{|\mathcal{K}_{\mathcal{S}}|}\mathbb{S}_{\mathcal{E}[:,k,:]}\right). \quad (3)$$

$G_n(\cdot)$ is still used to transform and decouple $\mathbb{S}_{\mathcal{E}}^{\mathcal{P}}$, obtaining more fundamental components of known classes. Then, after averaging these components, they are projected into the region representing "Unknown" in the feature space via a nonlinear transformation to generate a fake embedding prototype $\mathbb{F}_{\mathcal{E}}^{\mathcal{P}}$. The process is as follows:

$$\mathbb{F}_{\mathcal{E}}^{\mathcal{P}} = \text{MLP}\left(\frac{1}{|\mathcal{K}_{\mathcal{S}}|}\sum_{i=1}^{|\mathcal{K}_{\mathcal{S}}|}G_n\left(\mathbb{S}_{\mathcal{E}}^{\mathcal{P}}\right)_{[i,:]}\right). \quad (4)$$

After obtaining fake prototypes, two tasks need to be performed.

a) Main classification task: Calculate the distances between query samples and all prototypes to obtain the embedding classification loss $\mathcal{L}_{emb}^{dis}$. Specifically, cosine similarity is employed as the distance metric function to compute the embedding distance between $\mathbb{Q}_{\mathcal{E}}$ and $\mathbb{S}_{\mathcal{E}}^{\mathcal{P}} \cup \mathbb{F}_{\mathcal{E}}^{\mathcal{P}}$, as given by the following formula:

$$\mathcal{Q}_{emb}^{dis} = \tau_1\cdot\left(\frac{\mathbb{Q}_{\mathcal{E}}}{\|\mathbb{Q}_{\mathcal{E}}\|_2}\right)\cdot\left(\frac{\mathbb{S}_{\mathcal{E}}^{\mathcal{P}}\cup\mathbb{F}_{\mathcal{E}}^{\mathcal{P}}}{\left\|\mathbb{S}_{\mathcal{E}}^{\mathcal{P}}\cup\mathbb{F}_{\mathcal{E}}^{\mathcal{P}}\right\|_2}\right)^{\top} \quad (5)$$

where τ_1 is the learnable temperature parameter with an initial value of 10. $\mathcal{Q}_{emb}^{dis} \in \mathbb{R}^{\hat{B}\times(|\mathcal{K}_{\mathcal{S}}|+1)}$ contains distances $\mathcal{Q}_{\mathcal{K}_{emb}}^{dis}$ of known-class samples and distances $\mathcal{Q}_{\mathcal{U}_{emb}}^{dis}$ of unknown-class samples, and yields the loss through the cross-entropy function, which is elaborated in detail later.

b) Auxiliary contrastive learning task: Calculate the similarity scores between query samples and known-class prototypes to enhance intraclass compactness and interclass separability, thereby obtaining the embedding contrastive loss $\mathcal{L}_{emb}^{score}$. Similarly, contrastive learning is implemented by comparing the cosine similarity scores between $\mathbb{Q}_{\mathcal{E}}$ and $\mathbb{S}_{\mathcal{E}}^{\mathcal{P}}$, as

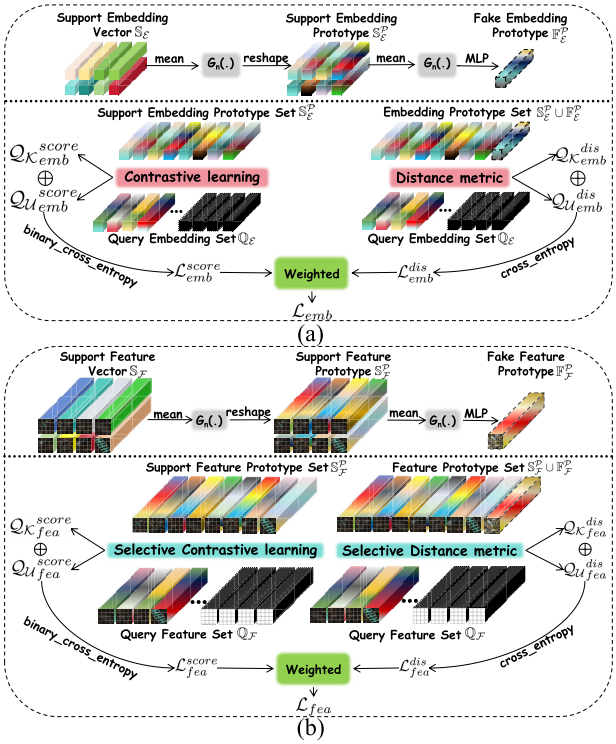


Fig. 4. (a) Schematic of the EWB. (b) Schematic of the FWB.

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follows:

$$\mathcal{Q}_{emb}^{score} = \tau_2 \cdot \left(\frac{\mathbb{Q}_{\mathcal{E}}}{\|\mathbb{Q}_{\mathcal{E}}\|_2} + 1\right) \cdot \left(\frac{\mathbb{S}_{\mathcal{E}}^{\mathcal{P}}}{\left\|\mathbb{S}_{\mathcal{E}}^{\mathcal{P}}\right\|_2} + 1\right)^{\top}. \quad (6)$$

τ_2 is set exactly the same as τ_1. Adding one aims to improve gradient quality and accelerate its convergence, and $\mathcal{Q}_{emb}^{score}$ also contains known-class scores $\mathcal{Q}_{\mathcal{K}emb}^{score}$ and unknown-class scores $\mathcal{Q}_{\mathcal{U}emb}^{score}$. With the aid of contrastive learning, known-class samples cluster near their class prototypes, while unknown-class samples are pushed away and drawn toward a fake prototype. If the distance of unknown samples to the fake prototype is the smallest, it is labeled "Unknown."

2) Feature-Wise Branch: In the FWB, the procedure for generating $\mathbb{S}_{\mathcal{F}}^{\mathcal{P}} \in \mathbb{R}^{|\mathcal{K}_{\mathcal{S}}|\times\hat{C}\times\hat{H}\times\hat{W}}$ and $\mathbb{F}_{\mathcal{F}}^{\mathcal{P}} \in \mathbb{R}^{1\times\hat{C}\times\hat{H}\times\hat{W}}$ is largely consistent with that in the EWB, except for an additional flattening step applied to the $\mathbb{S}_{\mathcal{F}}$. The process for both is described below

$$\begin{aligned}\mathbb{S}_{\mathcal{F}}^{\mathcal{P}} &= G_n\left(\frac{1}{|\mathcal{K}_{\mathcal{S}}|}\sum_{k=1}^{|\mathcal{K}_{\mathcal{S}}|}\text{Flatten}\left(\mathbb{S}_{\mathcal{E}}\right)_{[:,k,:]}\right)\\ \mathbb{F}_{\mathcal{F}}^{\mathcal{P}} &= \text{Reshape}\left(\text{MLP}\left(\frac{1}{|\mathcal{K}_{\mathcal{S}}|}\sum_{i=1}^{|\mathcal{K}_{\mathcal{S}}|}G_n\left(\mathbb{S}_{\mathcal{F}}^{\mathcal{P}}\right)_{[i,:]}\right)\right). \end{aligned} \quad (7)$$

Similarly, this branch also performs two tasks with a basic process consistent with EWB. However, in distance measurement and contrastive learning, it employs a top-$\hat{K}$ aggregation strategy, selecting the top-$\hat{K}$ most representative positions to enhance model robustness.

a) Main classification task: In this task, unlike EWB, a selective distance metric strategy is employed, selecting the top-$\hat{K}$ most relevant positions across prototype and query dimensions to enhance robustness and discriminability. $\mathbb{S}_{\mathcal{F}}^{\mathcal{P}} \cup \mathbb{F}_{\mathcal{F}}^{\mathcal{P}}$ and $\mathbb{Q}_{\mathcal{F}}$ are flattened along the spatial dimension as follows:

$$\begin{aligned}\left[\mathbb{S}_{\mathcal{F}}^{\mathcal{P}} \cup \mathbb{F}_{\mathcal{F}}^{\mathcal{P}}\right]_{\text{flat}} &= \text{Reshape}\left(\mathbb{S}_{\mathcal{F}}^{\mathcal{P}} \cup \mathbb{F}_{\mathcal{F}}^{\mathcal{P}}, \left[1, |\mathcal{K}_{\mathcal{S}}|+1, 1, \hat{C}, \hat{H}\hat{W}\right]\right)\\ \left[\mathbb{Q}_{\mathcal{E}}\right]_{\text{flat}} &= \text{Reshape}\left(\mathbb{Q}_{\mathcal{E}}, \left[\hat{B}, 1, \hat{H}\hat{W}, \hat{C}, 1\right]\right). \end{aligned} \quad (8)$$

Based on computing the cosine similarity between the two, the top-$\hat{K}$ most relevant positions are selected in the prototype and query dimensions, respectively, as follows:

$$\mathcal{Q}_{fea}^{dis} = \frac{1}{\hat{K}^2}\sum_{x=1}^{\hat{K}}\sum_{y=1}^{\hat{K}}\max_{k\in\text{query}}^{(x)}\max_{l\in\text{prototype}}^{(y)} \left[\left(\frac{\left[\mathbb{Q}_{\mathcal{F}}\right]_{\text{flat}}}{\left\|\left[\mathbb{Q}_{\mathcal{F}}\right]_{\text{flat}}\right\|_2}\right) \cdot \left(\frac{\left[\mathbb{S}_{\mathcal{F}}^{\mathcal{P}} \cup \mathbb{F}_{\mathcal{F}}^{\mathcal{P}}\right]_{\text{flat}}}{\left\|\left[\mathbb{S}_{\mathcal{F}}^{\mathcal{P}} \cup \mathbb{F}_{\mathcal{F}}^{\mathcal{P}}\right]_{\text{flat}}\right\|_2}\right)^{\top}\right]_{\text{dim}=3}_{i,j,k,l}. \quad (9)$$

$\mathcal{Q}_{fea}^{dis}$ alse contains distances $\mathcal{Q}_{\mathcal{K}fea}^{dis}$ and $\mathcal{Q}_{\mathcal{U}fea}^{dis}$.

b) Auxiliary contrastive learning task: In this task, selective contrastive learning is employed to enhance interclass discriminability. To improve complementarity, Euclidean distance is used as the scoring criterion. Consistent with the previous task, $\mathbb{S}_{\mathcal{F}}^{\mathcal{P}}$ and $\mathbb{Q}_{\mathcal{F}}$ are first flattened to obtain $[\mathbb{S}_{\mathcal{F}}^{\mathcal{P}}]_{\text{flat}}$ and $[\mathbb{Q}_{\mathcal{F}}]_{\text{flat}}$

$$\begin{aligned}\left[\mathbb{S}_{\mathcal{F}}^{\mathcal{P}}\right]_{\text{flat}} &= \text{Reshape}\left(\mathbb{S}_{\mathcal{F}}^{\mathcal{P}}, \left[1, |\mathcal{K}_{\mathcal{S}}|, \hat{C}, \hat{H}\hat{W}, 1\right]\right)\\ \left[\mathbb{Q}_{\mathcal{E}}\right]_{\text{flat}} &= \text{Reshape}\left(\mathbb{Q}_{\mathcal{E}}, \left[\hat{B}, 1, \hat{C}, 1, \hat{H}\hat{W}\right]\right). \end{aligned} \quad (10)$$

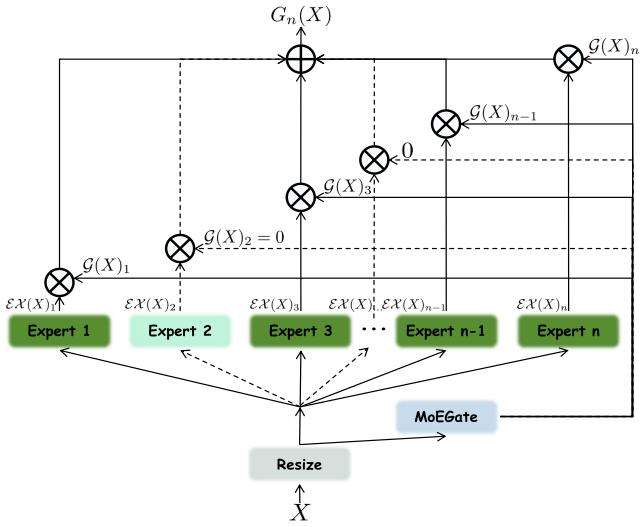


Fig. 5. Schematic illustration of $G_n(\cdot)$.

The selective learning process is as follows:

$$\mathcal{Q}_{fea}^{score} = 1 - \frac{1}{\hat{K}^2}\sum_{x=1}^{\hat{K}}\sum_{y=1}^{\hat{K}}\min_{k\in\text{query}}^{(x)}\min_{l\in\text{prototype}}^{(y)} \times\left[\left(\left\|\left[\mathbb{S}_{\mathcal{F}}^{\mathcal{P}}\right]_{\text{flat}} - \left[\mathbb{Q}_{\mathcal{F}}\right]_{\text{flat}}\right\|_2\right)_{\text{dim}=2}\right]_{i,j,k,l}. \quad (11)$$

$\mathcal{Q}_{fea}^{score}$ contains $\mathcal{Q}_{\mathcal{K}fea}^{score}$ and $\mathcal{Q}_{\mathcal{U}fea}^{score}$. $\hat{K}$ is set to 5.

$G_n(\cdot)$ employs the mixture-of-experts (MoE) architecture, where each expert is implemented with a multihead attention mechanism, with the process illustrated in Fig. 5. MoEGate serves as a gating unit, which receives inputs, determines which experts process each input sample, and generates their routing weights. It greatly enhances the richness of model representation under controllable cost. Specifically, for the ith expert, its output is defined as

$$\mathcal{EX}(X)_i = \text{LayerNorm}_i\left(\sigma\left(\text{Attention}_i(X)\right) + X\right) \quad (12)$$

where σ denotes the softmax function. MoEGate is implemented using an MLP to compute expert scores, activating the top 50% of experts by score with the scores of the remaining experts set to 0, as follows:

$$\mathcal{G}(X)_i = \begin{cases}\sigma\left(\text{MLP}(X)_i\right), & i \text{ is in the top 50\% position}\\ 0, & \text{otherwise.}\end{cases} \quad (13)$$

Thus, the final output is given by

$$G_n(X) = \text{LayerNorm}\left(\sum_{i=1}^{n}\mathcal{G}(X)_i\,\mathcal{EX}(X)_i + X\right). \quad (14)$$

3) Energy-Based Module: The EBM employs an energy loss mechanism, which combines and weights the metric distances ($\mathcal{Q}_{\mathcal{K}emb}^{dis}, \mathcal{Q}_{\mathcal{U}emb}^{dis}, \mathcal{Q}_{\mathcal{K}fea}^{dis}, \mathcal{Q}_{\mathcal{U}fea}^{dis}$) from EWB and FWB. This ensures known-class samples have energy lower than the threshold m_{in} while unknown-class samples have energy higher than the threshold m_{out}, thus enhancing OSR performance. First, energy scores are computed for the four metric distances, as follows:

$$E_{\mathcal{K}}^{\mathcal{E}} = -\log\sum_{i=1}^{|\mathcal{K}_{\mathcal{S}}|}\exp\left(\left[\mathcal{Q}_{\mathcal{K}emb}^{dis}\right]_i\right)$$

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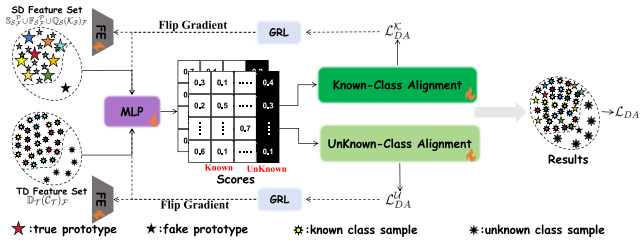


Fig. 6. Schematic illustration of the decoupled adversarial DA module.

$$E_{\mathcal{U}}^{\mathcal{E}} = -\log \sum_{i=1}^{|\mathcal{K}_{\mathcal{S}}|} \exp\left(\left[\mathcal{Q}_{\mathcal{U}emb}^{dis}\right]_i\right)$$
$$E_{\mathcal{K}}^{\mathcal{F}} = -\log \sum_{i=1}^{|\mathcal{K}_{\mathcal{S}}|} \exp\left(\left[\mathcal{Q}_{\mathcal{K}fea}^{dis}\right]_i\right)$$
$$E_{\mathcal{U}}^{\mathcal{F}} = -\log \sum_{i=1}^{|\mathcal{K}_{\mathcal{S}}|} \exp\left(\left[\mathcal{Q}_{\mathcal{U}fea}^{dis}\right]_i\right). \tag{15}$$

Fuse the energy scores of known classes and unknown classes separately, then compute the energy loss, as follows:

$$\mathcal{L}_{energy} = \alpha \cdot \left(\mathbb{E}_{\mathcal{K}}\left[\left(\mathrm{ReLU}\left(E_{\mathcal{K},i}^{final} - m_{\mathrm{in}}\right)\right)^2\right] + \mathbb{E}_{\mathcal{U}}\left[\left(\mathrm{ReLU}\left(m_{\mathrm{out}} - E_{\mathcal{U},j}^{final}\right)\right)^2\right]\right) \tag{16}$$

where α, m_{in}, and m_{out} are all learnable parameters, with initial values of 0.1, -1, and 1. Besides, $E_{\mathcal{K}}^{final} = E_{\mathcal{K}}^{\mathcal{E}} + E_{\mathcal{K}}^{\mathcal{F}}$, and $E_{\mathcal{U}}^{final} = E_{\mathcal{U}}^{\mathcal{E}} + E_{\mathcal{U}}^{\mathcal{F}}$.

D. Decoupled Adversarial DA Module

A decoupled adversarial strategy achieves DA between all SD samples and known TD samples, while rejecting unknown TD classes. Since the SD contains no unknown classes and its label space is entirely disjoint from the TD, fake feature prototypes $\mathbb{F}_{\mathcal{F}}^{\mathcal{P}}$ are introduced to represent unknown classes in the SD. This strategy separately aligns known and unknown classes across domains, as illustrated in Fig. 6. The data used for DA in the SD is $\mathbb{S}_{\mathcal{S}\mathcal{F}}^{\mathcal{P}} \cup \mathbb{F}_{\mathcal{S}\mathcal{F}}^{\mathcal{P}} \cup \mathbb{Q}_{\mathcal{S}}(\mathcal{K}_{\mathcal{S}})_{\mathcal{F}}$, while that in the TD is $\mathbb{D}_{\mathcal{T}}(\mathcal{C}_{\mathcal{T}})_{\mathcal{F}}$. Compared with embedding vectors, feature vectors possess richer spatial features and can provide more information for DA. For simplicity, the sample size in SD is set to $B_{\mathcal{S}}$, while that in TD is set to $B_{\mathcal{T}}$. Specifically, after flattening the SD/TD features, the MLP is used to map them, followed by a softmax to obtain confidence scores for classes. Subsequently, the confidence of each sample belonging to known or unknown classes is calculated. The process is as follows:

$$\mathcal{W}_{\mathcal{K}}^{\mathcal{S}/\mathcal{T}}{}_i = \sum_{c=1}^{|\mathcal{C}_{\mathcal{T}}^{\mathcal{K}}|} \left[\sigma\left(\mathrm{MLP}\left(\mathrm{Flatten}\left(X^{\mathcal{S}/\mathcal{T}}\right)\right)\right)\right]_{i,c}$$
$$\mathcal{W}_{\mathcal{U}}^{\mathcal{S}/\mathcal{T}}{}_i = \left[\sigma\left(\mathrm{MLP}\left(\mathrm{Flatten}\left(X^{\mathcal{S}/\mathcal{T}}\right)\right)\right)\right]_{i,|\mathcal{C}_{\mathcal{T}}^{\mathcal{K}}|+1}. \tag{17}$$

Subsequently, two domain discriminators $f_{\mathcal{K}}$ and $f_{\mathcal{U}}$ are designed based on the gradient reversal layer (GRL) of the adversarial strategy, which are used for the alignment of known classes and unknown classes between the SD and TD, respectively. $\mathcal{L}_{DA}^{\mathcal{K}}$ and $\mathcal{L}_{DA}^{\mathcal{U}}$ are obtained, respectively, based on the binary cross-entropy loss, as follows:

$$\mathcal{L}_{DA}^{\mathcal{K}} = -\frac{1}{B_{\mathcal{S}}}\sum_{i=1}^{B_{\mathcal{S}}} \mathcal{W}_{\mathcal{K}i}^{\mathcal{S}} \cdot y_i^{\mathcal{S}} \cdot \log\left(f_{\mathcal{K}}\left(x_i^{\mathcal{S}}\right)\right) - \frac{1}{B_{\mathcal{T}}}\sum_{i=1}^{B_{\mathcal{T}}} \mathcal{W}_{\mathcal{K}i}^{\mathcal{T}} \cdot \left(1 - y_i^{\mathcal{T}}\right) \cdot \log\left(f_{\mathcal{K}}\left(x_i^{\mathcal{T}}\right)\right)$$
$$\mathcal{L}_{DA}^{\mathcal{U}} = -\frac{1}{B_{\mathcal{S}}}\sum_{i=1}^{B_{\mathcal{S}}} \mathcal{W}_{\mathcal{U}i}^{\mathcal{S}} \cdot y_i^{\mathcal{S}} \cdot \log\left(f_{\mathcal{U}}\left(x_i^{\mathcal{S}}\right)\right) - \frac{1}{B_{\mathcal{T}}}\sum_{i=1}^{B_{\mathcal{T}}} \mathcal{W}_{\mathcal{U}i}^{\mathcal{T}} \cdot \left(1 - y_i^{\mathcal{T}}\right) \cdot \log\left(f_{\mathcal{U}}\left(x_i^{\mathcal{T}}\right)\right) \tag{18}$$

where y_i denotes the domain label of the ith sample, while x_i denotes the flattened ith sample. DA loss $\mathcal{L}_{DA}$ is the sum of the two, i.e., $\mathcal{L}_{DA} = \mathcal{L}_{DA}^{\mathcal{K}} + \mathcal{L}_{DA}^{\mathcal{U}}$. Notably, the class spaces in SD and TD are completely inconsistent. Thus, there is no strong correlation in the alignment of their known-class features. This results in the initial DA being only a rough alignment. As SD and TD perform alternate optimization of HVSE, the classification loss acts back on the DA process, making the weights of known classes in SD and TD more prominent, thereby gradually achieving precise alignment.

E. Loss Function

The loss function consists of $\mathcal{L}_{emb}^{\mathcal{S}/\mathcal{T}}$, $\mathcal{L}_{fea}^{\mathcal{S}/\mathcal{T}}$, $\mathcal{L}_{energy}^{\mathcal{S}/\mathcal{T}}$, and $\mathcal{L}_{DA}$ four components, among which $\mathcal{L}_{energy}^{\mathcal{S}/\mathcal{T}}$ and $\mathcal{L}_{DA}$ have been elaborated on in previous sections. Taking the SD as the example, $\mathcal{L}_{emb}$ is obtained by adding $\mathcal{L}_{emb}^{dis}$ and $\mathcal{L}_{emb}^{score}$, which are obtained in the following ways:

$$\mathcal{L}_{emb}^{dis} = -\frac{1}{\hat{B}}\sum_{i=0}^{\hat{B}-1}\left[\left[\mathcal{Q}_{emb}^{dis}\right]_{i,y_i} - \log\sum_{y'=1}^{|\mathcal{K}_{\mathcal{S}}|+1}\exp\left(\left[\mathcal{Q}_{emb}^{dis}\right]_{i,y'}\right)\right]$$
$$\mathcal{L}_{emb}^{score} = -\sum_{i=1}^{|\mathcal{K}_{\mathcal{S}}|}\sum_{q_j\in\mathcal{K}_{\mathcal{S}}}\log\left(\theta\left(\left[\mathcal{Q}_{\mathcal{K}emb}^{score}\right]_{i,q_j}\right)\right) - \sum_{i=1}^{|\mathcal{K}_{\mathcal{S}}|}\sum_{q_j\in\mathcal{U}_{\mathcal{T}}}\log\left(1-\theta\left(\left[\mathcal{Q}_{\mathcal{U}emb}^{score}\right]_{i,q_j}\right)\right). \tag{19}$$

The $\mathcal{L}_{fea}^{dis}$ and $\mathcal{L}_{fea}^{score}$ of FWB are obtained in the same manner and then merged with the same weights. Due to the peculiarity of DA, DA is performed only when FSL is conducted on the SD. Hence, $\mathcal{L}_{\mathrm{DA}}$ emerges solely in the SD. In addition, prior work has demonstrated the efficacy of conjugate tasks, so conjugate training is incorporated into CDFSOSR. The total losses on SD and TD are formulated as follows, respectively, where $o = p$ and $o = c$ denote the primary task and conjugate task, respectively,

$$\mathcal{L}^{\mathcal{S}} = \mathcal{L}_{DA} + \sum_{o}^{\{p,c\}}\left((1-\lambda)\mathcal{L}_{emb}^{\mathcal{S}} + \lambda\mathcal{L}_{fea}^{\mathcal{S}} + \lambda\mathcal{L}_{energy}^{\mathcal{S}}\right)_o$$
$$\mathcal{L}^{\mathcal{T}} = \sum_{o}^{\{p,c\}}\left((1-\lambda)\mathcal{L}_{emb}^{\mathcal{T}} + \lambda\mathcal{L}_{fea}^{\mathcal{T}} + \lambda\mathcal{L}_{energy}^{\mathcal{T}}\right)_o. \tag{20}$$

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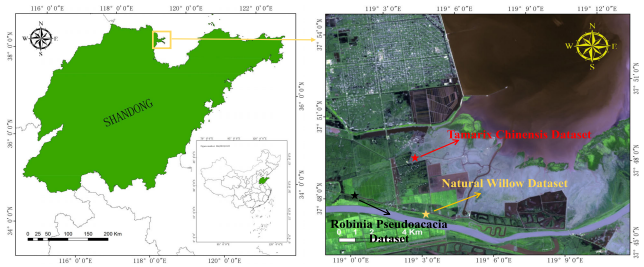


Fig. 7. Overview of the study area.

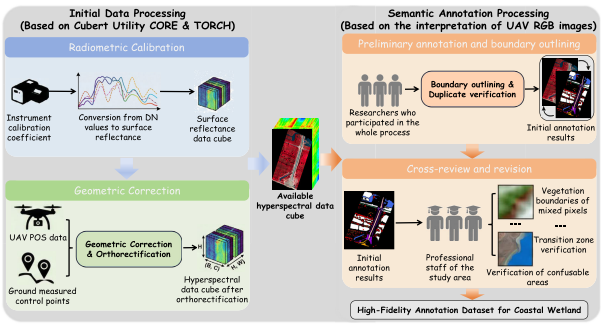


Fig. 8. Processing workflow for the three self-constructed datasets.

IV. Experiments and Discussion

A. Area of Study and Datasets

1) Yellow River Delta: The study area is located within the Yellow River Delta National Nature Reserve in Shandong, China, characterized by its diverse vegetation, such as robinia, tamarisks, willows, and reeds. Three subregions were selected for monitoring wetland vegetation coverage using HSIC, as shown in Fig. 7 [37].

2) SD Dataset: Derived from the Chikusei (CKS) HSI acquired in Ibaraki, Japan, the SD dataset has 128 bands, 2.5-m resolution, 19 classes, and dimensions of 2517×2335 pixels [38].

3) TD Datasets: Five TD datasets are used, including three self-constructed wetland vegetation sets from the Yellow River Delta and two public datasets.

1) *NW Dataset:* Focusing on natural willow (NW) forests, the NW dataset was captured in September 2022 via a DJI M600 Pro UAV equipped with a Cubert Utilities Firefly S185 camera. The dataset has dimensions of $590 \times 512 \times 126$, a spectral range of 0.4–0.95 μm, and a spatial resolution of 1 m.
2) *TC Dataset:* Processed consistently with the NW dataset, the *Tamarix chinensis* (TC) dataset was captured on September 25, 2022. Focusing on tamarisk forests, it contains nine land cover types, has dimensions of $784 \times 1024 \times 126$, and a spatial resolution of 0.5 m.
3) *RP Dataset:* Acquired on June 21, 2023, at 11:15 via a DJI UAV, the *Robinia pseudoacacia* (RP) dataset targets RP forests. It contains 11 land cover types, has a size of 1484×1246 pixels, 126 spectral bands, and a spatial resolution of 1 m.
4) *PU Dataset:* Acquired by ROSIS over the University of Pavia, the Pavia University (PU) dataset has an image size of 610×340 pixels, a spatial resolution of 1.3 m, and 103 spectral bands. It includes nine land cover classes and comprises 42 776 labeled samples.
5) *YC Wetlands Dataset:* Spanning the coastal areas of Yancheng, Jiangsu, China, the Yancheng (YC) dataset has a spatial resolution of 30 m, was captured by the AHSI on the ZY1-02D satellite, and has dimensions of $1175 \times 585 \times 253$. It contains 18 land cover types and 7894 labeled samples.

The ground truth maps, number of categories, land cover types, and the number of labels for each type across the six datasets are presented in Table I, where the shaded categories and white represent the unknown class in TD.

B. Data Preprocessing

The processing workflow for the three self-constructed datasets (NW, TC, and RP) is described in Fig. 8. First, full radiometric calibration and geometric correction were performed via Cubert's official proprietary software, Cubert Utility CORE and TORCH. Radiometric calibration converted digital number (DN) values to surface reflectance (56%, 32%, and 11% calibration references) using instrument-specific coefficients, while geometric correction was implemented with uncrewed aerial vehicle (UAV) position and orientation system (POS) data and field-measured ground control points (GCPs).

Subsequently, a two-stage annotation was conducted with UAV RGB imagery as the interpretation reference. In Stage 1, three researchers with full field survey experience and thorough familiarity with the study area's land cover characteristics completed preliminary land cover annotation and boundary delineation. In Stage 2, cross-validation and inter-annotator agreement assessment were performed: Yellow River Delta Reserve administrators independently cross-checked and revised preliminary annotations, focusing on correcting errors in ambiguous regions (e.g., mixed-pixel dense vegetation boundaries, tidal creek-water transition zones), to generate the final high-fidelity coastal wetland annotated dataset.

C. Experimental Setting

1) Evaluation Metrics: The performance of DBEDML is evaluated by overall accuracy (OA), average accuracy (AA), Kappa coefficient (Kappa), and the accuracy of the "Unknown" class.

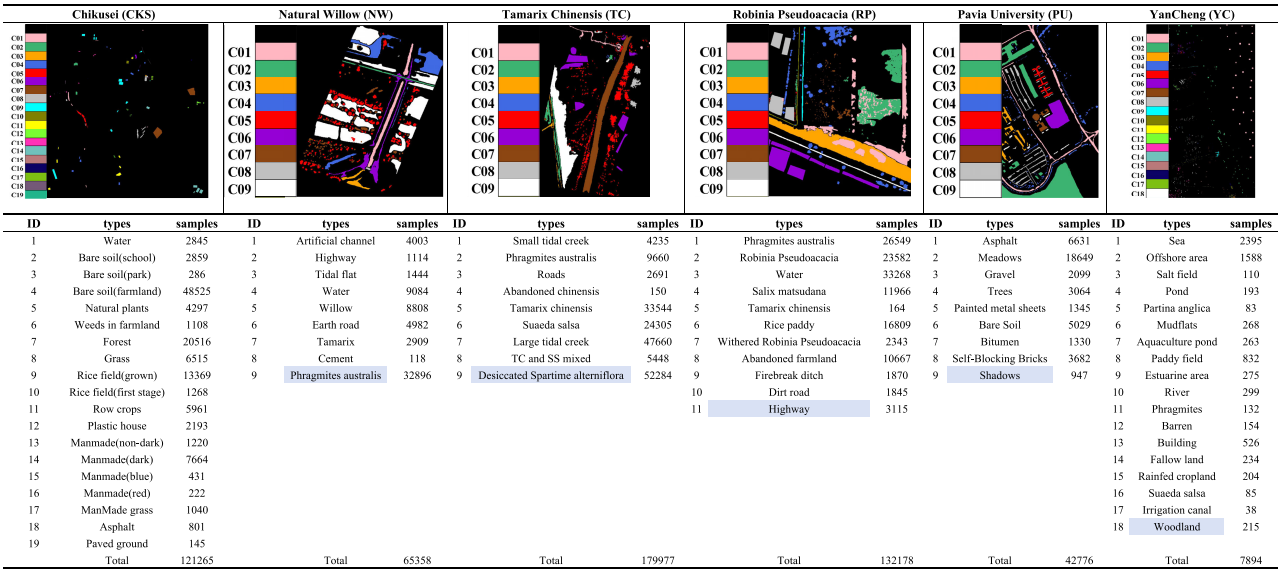
2) Comparison Algorithms: To evaluate the performance of the proposed DBEDML, it is compared with nine state-of-the-art algorithms across different focus domains. In supervised learning HSIC, we use SSFTT [39], CSCANet [40], and the foundation model HyperSIGMA [41]. In the CDFSL HSIC domain, CMFSL [42] and FDFSL [43] are adopted. For the aforementioned algorithms, an additional Softmax function is added based on the original algorithms to score each sample. If the maximum score across all classes is less than 0.5, the sample is classified as "Unknown." For OSR HSIC, OSC-SCL [44], OSC_OCL [45], SSMLP-RPL [46], and HyperTaFOR [20] are used for comparison. All parameter settings are consistent with those in the original papers.

3) Configuration of Hyperparameters: DBEDML is implemented using Python 3.10, CUDA 11.2, and Torch 1.10, with

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TABLE I

GROUND TRUTH AND THE SPECIFIC SAMPLE COUNTS FOR EACH CLASS IN BOTH SD AND TD DATASETS



ID	types	samples	ID	types	samples	ID	types	samples	ID	types	samples	ID	types	samples	ID	types	samples
1	Water	2845	1	Artificial channel	4003	1	Small tidal creek	4235	1	Phragmites australis	26549	1	Asphalt	6631	1	Sea	2395
2	Bare soil(school)	2859	2	Highway	1114	2	Phragmites australis	9660	2	Robinia Pseudoacacia	23582	2	Meadows	18649	2	Offshore area	1588
3	Bare soil(park)	286	3	Tidal flat	1444	3	Roads	2691	3	Water	33268	3	Gravel	2099	3	Salt field	110
4	Bare soil(farmland)	48525	4	Water	9084	4	Abandoned chinensis	150	4	Salix matsudana	11966	4	Trees	3064	4	Pond	193
5	Natural plants	4297	5	Willow	8808	5	Tamarix chinensis	33544	5	Tamarix chinensis	164	5	Painted metal sheets	1345	5	Partina anglica	83
6	Weeds in farmland	1108	6	Earth road	4982	6	Suaeda salsa	24305	6	Rice paddy	16809	6	Bare Soil	5029	6	Mudflats	268
7	Forest	20516	7	Tamarix	2909	7	Large tidal creek	47660	7	Withered Robinia Pseudoacacia	2343	7	Bitumen	1330	7	Aquaculture pond	263
8	Grass	6515	8	Cement	118	8	TC and SS mixed	5448	8	Abandoned farmland	10667	8	Self-Blocking Bricks	3682	8	Paddy field	832
9	Rice field(grown)	13369	9	Phragmites australis	32896	9	Desiccated Spartime alterniflora	52284	9	Firebreak ditch	1870	9	Shadows	947	9	Estuarine area	275
10	Rice field(first stage)	1268							10	Dirt road	1845				10	River	299
11	Row crops	5961							11	Highway	3115				11	Phragmites	132
12	Plastic house	2193													12	Barren	154
13	Manmade(non-dark)	1220													13	Building	526
14	Manmade(dark)	7664													14	Fallow land	234
15	Manmade(blue)	431													15	Rainfed cropland	204
16	Manmade(red)	222													16	Suaeda salsa	85
17	ManMade grass	1040													17	Irrigation canal	38
18	Asphalt	801													18	Woodland	215
19	Paved ground	145															
	Total	121265		Total	65358		Total	179977		Total	132178		Total	42776		Total	7894

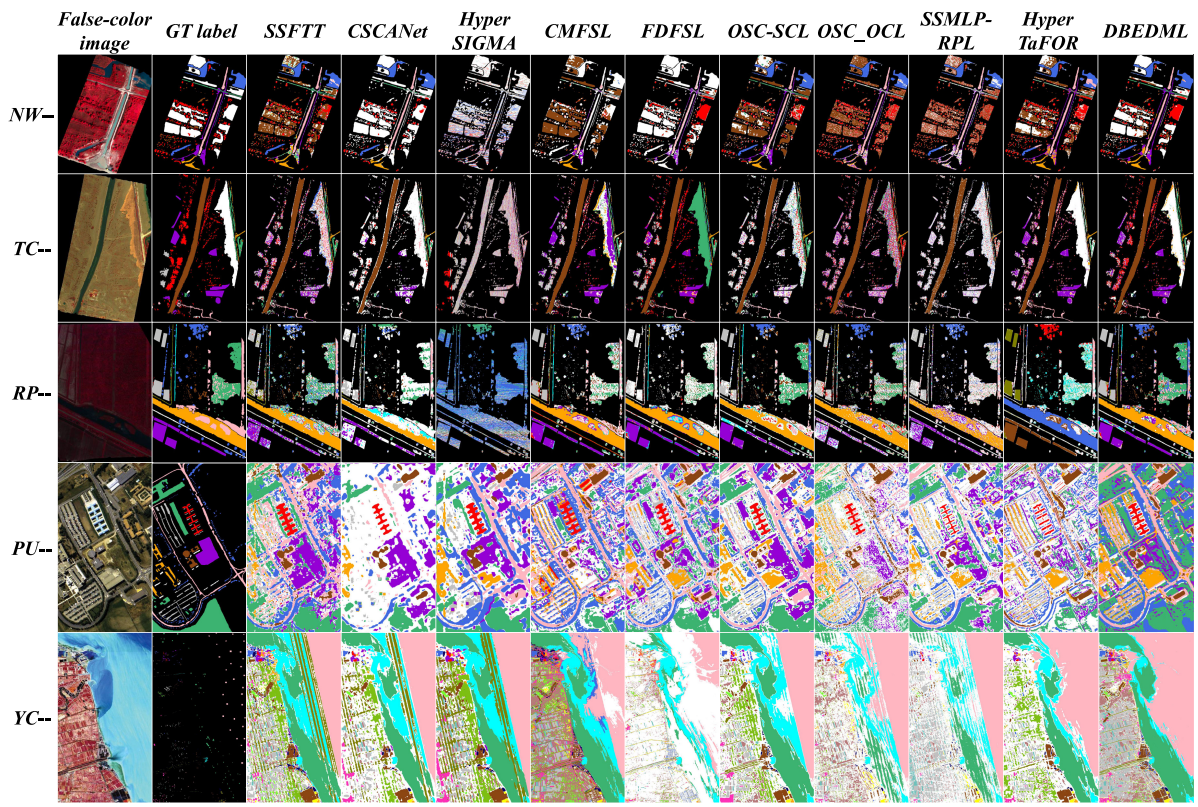


Fig. 9. False color maps, ground truth, and classification maps of ten algorithms for five datasets.

training conducted on a computer equipped with an NVIDIA RTX 3090 GPU, 96 GB of RAM, and an Intel Core i9-10900K CPU. Results are independently conducted ten times, reported as "mean ± standard deviation."

D. Experimental Results

The classification accuracies of the ten algorithms on the five datasets are presented in Table II(a)–(e), respectively. The results highlighted in bold represent the optimal results. The false-color images, ground truth maps, and classification maps of each algorithm are shown in Fig. 9.

Table II shows that DBEDML consistently outperforms all algorithms in OA, AA, Kappa, and "Unknown" class accuracy across five datasets, demonstrating strong generalization and stability. It surpasses the next best methods by 16.16%, 10.63%, 8.58%, 4.82%, and 3.26%, respectively. Algorithms in the CDFSL HSIC and supervised learning HSIC domains generally perform poorly, with lower OA and AA despite

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TABLE II

COMPARISON OF CLASSIFICATION RESULTS FOR TEN METHODS ON THE FIVE DATASETS (FIVE LABELED SAMPLES PER CLASS)

	Class	SSFTT*	CSCANet*	HyperSIGMA*	CMFSL*	FDFSL*	OSC-SCL	OSC_OCL	SSMLP-RPL	HyperTaFOR	DBEDML
(a) NW	1	**99.37±0.61**	76.18±40.21	82.21±17.35	19.81±19.88	39.50±38.61	91.98±2.90	95.59±3.27	93.94±4.73	98.95±1.69	97.42±2.83
	2	74.36±10.87	35.31±26.86	65.85±13.15	16.27±11.17	27.09±29.39	**92.12±9.16**	86.04±10.55	88.72±11.38	79.54±20.07	86.74±14.47
	3	92.20±5.96	71.08±27.17	**95.92±4.32**	18.99±24.56	36.87±38.91	93.65±6.86	85.53±18.12	80.99±19.41	95.07±3.91	86.57±6.61
	4	90.27±3.45	79.56±10.64	89.57±3.59	7.37±22.04	49.66±40.89	**92.29±7.01**	86.76±6.39	85.87±9.34	91.84±4.05	89.46±5.98
	5	72.72±8.85	51.60±15.11	44.60±13.08	5.42±16.18	**87.91±22.35**	48.47±12.91	52.51±13.75	52.29±15.53	75.30±11.26	63.22±24.03
	6	56.15±13.03	34.56±16.04	51.07±14.47	35.82±27.76	45.34±34.57	70.77±9.97	67.55±15.95	74.96±16.76	78.33±15.65	**86.62±10.09**
	7	**73.65±9.12**	42.60±30.96	53.28±10.10	2.67±6.53	12.02±22.72	72.87±16.08	61.77±10.90	67.30±14.57	52.41±15.46	42.52±20.82
	8	**89.65±5.97**	84.34±8.81	75.49±14.86	21.59±32.36	23.98±24.69	82.30±6.74	65.22±11.87	67.97±15.66	60.00±10.73	87.08±7.36
	Unknown	8.49±7.89	47.83±37.00	25.22±8.09	49.78±0.36	18.78±27.64	13.42±9.80	17.07±18.21	13.14±11.41	42.01±14.39	**77.81±11.31**
	OA(%)	43.70±3.70	53.6±19.28	45.81±1.88	54.27±4.43	61.43±7.41	44.79±4.14	44.94±8.79	43.46±4.16	61.97±6.78	**78.13±3.94**
	AA(%)	72.98±1.98	58.12±4.74	64.80±3.43	19.75±10.68	37.91±12.18	73.00±2.20	66.29±5.17	69.46±3.75	74.83±4.07	**79.72±4.75**
	Kappa(*100)	37.25±2.79	39.53±20.20	35.96±2.88	10.83±11.79	34.52±15.23	36.46±3.28	36.43±8.18	35.34±3.10	53.68±6.57	**69.88±4.50**
(b) TC	1	51.91±10.59	56.41±21.99	46.71±17.36	20.55±19.04	48.55±27.90	**74.27±11.88**	68.46±4.80	50.00±21.87	55.24±14.87	64.74±10.51
	2	76.49±12.20	72.66±7.70	**80.36±8.42**	27.77±12.34	44.55±27.85	61.98±12.51	76.93±7.89	61.52±14.81	39.39±16.20	61.76±11.95
	3	82.86±11.70	77.96±29.83	89.04±6.96	43.02±28.55	66.71±34.68	82.52±13.44	83.32±5.59	74.43±10.15	**89.29±6.74**	83.74±11.00
	4	89.66±10.98	**93.24±7.83**	**93.24±7.80**	34.21±27.36	60.90±32.94	74.41±23.75	74.19±13.72	74.55±8.32	62.83±14.21	85.03±9.68
	5	23.40±8.52	25.63±13.64	20.51±16.45	14.31±14.43	38.93±23.93	10.62±4.81	32.19±14.53	18.48±7.80	27.78±6.48	**50.26±9.24**
	6	39.82±8.06	44.12±18.81	45.35±12.95	34.70±28.73	45.96±23.20	23.56±14.73	35.40±6.28	29.70±9.15	68.11±13.23	**69.81±13.68**
	7	96.04±1.90	86.76±30.54	96.61±2.01	63.25±39.58	91.66±7.65	92.81±4.33	92.06±3.99	93.39±4.06	96.36±1.49	**97.02±2.45**
	8	41.84±9.90	**51.63±13.99**	38.38±12.79	38.13±29.78	43.83±26.42	44.39±12.28	17.33±4.24	26.14±13.03	34.09±11.48	41.44±16.39
	Unknown	41.54±9.07	57.91±15.68	46.34±13.64	70.79±10.65	76.61±30.82	63.97±15.16	33.79±4.90	39.37±24.87	81.87±16.95	**97.20±2.56**
	OA(%)	55.14±3.99	58.57±11.29	56.97±6.34	53.21±12.62	65.90±10.64	56.03±4.16	46.64±5.31	50.07±5.48	69.51±6.14	**80.14±2.59**
	AA(%)	60.39±4.23	62.92±5.40	60.47±3.60	38.52±9.14	57.52±16.38	58.73±3.86	57.07±3.39	51.95±3.36	57.02±5.03	**72.33±2.33**
	Kappa(*100)	45.21±4.76	48.3±13.57	47.99±7.80	37.23±18.33	56.35±14.03	44.01±4.72	33.65±5.16	37.70±5.26	61.45±7.40	**74.93±3.19**
(c) RP	1	20.46±5.12	24.19±7.85	14.25±6.17	5.31±12.73	26.11±15.70	16.92±6.52	17.29±8.27	18.54±5.74	21.93±6.99	**30.50±6.96**
	2	35.08±11.09	46.47±10.49	56.19±11.56	14.46±26.92	41.07±29.66	55.07±19.42	29.95±15.06	23.14±14.83	33.74±10.61	**63.84±11.18**
	3	84.53±9.15	87.35±7.91	91.59±3.84	75.87±23.99	75.99±28.82	85.40±5.49	76.05±24.83	84.60±13.22	**96.82±1.58**	95.91±2.56
	4	40.66±10.65	42.50±8.41	30.80±13.04	11.19±11.90	29.26±22.78	41.62±8.52	47.53±12.58	40.04±14.33	40.03±13.55	**48.93±11.93**
	5	39.94±9.03	46.23±8.20	66.48±10.57	9.25±15.19	40.00±35.77	52.01±14.04	30.57±16.18	37.86±15.85	55.85±8.15	**71.13±9.30**
	6	68.11±10.96	83.02±10.31	**98.11±1.63**	20.62±28.68	57.02±33.51	72.43±7.11	48.33±17.09	53.33±19.61	86.96±7.17	88.43±7.74
	7	69.98±9.85	75.51±7.65	**84.98±3.92**	20.11±24.76	41.39±33.57	53.09±16.81	24.61±14.46	30.69±17.52	60.41±20.49	64.56±14.34
	8	58.71±12.22	54.07±23.14	75.20±15.80	38.58±31.27	58.20±26.96	89.31±6.13	35.66±33.80	77.21±15.76	**91.68±6.46**	89.63±6.62
	9	34.82±9.89	25.60±19.04	38.64±18.27	19.39±25.78	32.47±32.45	45.96±13.36	29.80±11.18	43.74±18.73	60.18±10.20	**64.70±18.25**
	10	58.10±6.52	48.55±12.91	28.36±11.32	24.33±29.68	54.04±31.13	54.48±16.54	41.96±17.33	46.44±21.66	71.81±11.57	**80.38±8.95**
	Unknown	32.51±17.24	58.41±16.13	49.13±11.59	83.00±25.46	47.55±41.29	50.39±24.57	77.64±25.65	68.09±27.89	50.39±25.00	**89.44±29.82**
	OA(%)	52.08±3.55	57.71±2.87	60.96±2.91	32.43±12.05	49.41±18.94	58.70±4.79	44.60±8.40	49.25±4.14	61.06±2.78	**69.94±3.17**
	AA(%)	49.35±2.90	53.81±3.44	54.89±1.77	29.28±10.88	45.74±17.40	56.63±4.66	40.60±5.22	47.61±4.88	60.89±3.09	**71.59±5.29**
	Kappa(*100)	45.75±3.53	51.83±3.01	55.29±3.79	26.79±11.80	43.75±18.85	53.22±4.92	38.42±8.07	43.16±4.14	55.60±2.94	**64.83±3.62**
(d) PU	1	70.62±8.92	24.48±14.27	25.00±12.18	27.86±22.23	**73.50±12.89**	68.22±12.67	59.31±16.11	55.79±21.29	42.94±16.89	72.21±12.13
	2	37.95±16.92	47.08±14.88	**74.33±11.11**	39.68±28.16	71.90±12.16	60.41±14.24	70.97±17.06	67.26±19.60	46.07±10.37	72.83±9.88
	3	71.82±13.59	39.36±18.14	71.10±16.25	20.49±25.74	49.57±26.25	**75.49±13.81**	70.31±9.42	62.96±13.46	56.76±10.72	66.85±4.76
	4	74.96±8.59	54.34±12.19	36.69±15.80	39.67±30.71	72.01±23.35	82.90±5.68	76.90±9.23	78.56±6.06	77.28±8.36	**84.01±10.26**
	5	**100.00±0.00**	95.95±2.67	99.72±0.75	75.54±33.18	62.24±44.05	99.16±0.95	94.86±5.96	92.67±7.51	84.62±7.57	97.40±4.00
	6	**95.48±11.87**	49.21±21.30	74.94±15.98	18.46±24.84	58.15±22.76	82.91±14.75	65.74±16.15	64.06±22.84	40.04±11.00	58.25±13.52
	7	**99.02±1.52**	64.43±35.00	86.88±9.59	46.85±30.73	67.92±31.85	91.17±6.27	73.50±12.01	76.02±13.10	81.20±6.18	83.93±6.40
	8	**64.84±16.84**	14.27±8.84	26.48±11.41	25.35±19.61	65.41±30.12	75.87±21.37	50.56±22.24	54.75±16.71	51.64±13.68	61.72±12.48
	Unknown	5.07±12.50	85.60±16.97	24.94±4.97	64.34±32.83	21.56±29.95	45.75±30.45	25.40±20.11	48.94±37.42	81.23±22.74	**98.12±2.02**
	OA(%)	59.51±6.07	44.07±7.27	59.88±6.32	35.07±11.55	67.35±8.45	67.43±4.50	67.01±6.99	65.29±7.05	55.96±5.22	**72.25±4.06**
	AA(%)	68.86±2.37	52.74±5.10	57.79±4.25	39.80±7.85	60.25±12.64	75.76±4.19	65.28±4.72	66.78±5.14	68.16±3.11	**77.26±2.59**
	Kappa(*100)	52.23±6.47	36.24±6.22	50.78±7.19	27.32±8.87	59.17±10.06	64.12±6.90	59.41±7.32	57.57±7.11	48.02±5.10	**64.89±4.37**
(e) YC	1	87.88±9.31	86.02±10.03	94.00±5.28	55.05±38.93	51.18±42.21	96.07±3.49	92.45±3.40	87.53±12.59	**99.77±0.55**	99.28±1.65
	2	93.13±4.20	93.15±4.48	93.37±4.40	45.43±45.52	40.11±41.23	98.00±3.42	92.88±6.14	94.33±5.93	**99.93±0.12**	99.92±0.15
	3	76.67±13.08	71.62±10.85	71.24±17.19	16.38±26.03	30.10±38.20	78.67±9.22	68.28±21.84	69.43±11.80	52.67±14.23	**85.81±8.48**
	4	69.79±12.30	67.39±12.00	58.67±10.10	25.64±22.28	16.33±29.65	68.94±16.55	57.45±17.42	60.05±14.73	62.61±8.20	**71.01±9.67**
	5	**99.49±1.62**	98.21±3.21	89.36±14.03	22.69±33.54	47.82±46.72	98.59±4.46	88.97±14.61	89.74±7.62	98.08±2.79	**99.49±0.85**
	6	**87.30±5.21**	76.01±7.42	52.55±12.40	33.61±33.76	49.85±32.37	80.87±7.25	64.75±18.97	66.62±9.17	77.80±9.64	83.95±9.17
	7	81.86±4.10	73.80±7.49	66.44±17.35	22.21±32.17	20.00±34.67	85.50±6.51	52.71±19.04	51.59±20.31	83.72±8.54	**87.25±8.03**
	8	88.69±7.26	74.26±10.16	47.88±10.92	50.45±32.13	30.17±33.25	90.19±3.41	81.21±8.96	83.96±13.28	98.07±1.29	**98.21±1.42**
	9	89.08±5.47	84.59±9.25	91.22±6.05	47.37±31.61	38.22±38.90	90.37±7.87	82.11±12.95	80.48±17.81	87.11±6.59	**91.30±5.09**
	10	44.73±14.24	33.16±9.10	46.80±7.59	19.01±21.74	12.59±19.91	72.35±9.30	50.34±16.55	42.25±16.58	83.71±8.61	**91.09±5.02**
	11	93.23±12.03	89.37±12.01	77.72±15.14	22.68±33.21	43.07±42.18	91.10±12.64	86.22±14.61	88.98±11.94	86.93±9.94	**95.51±8.85**
	12	95.10±5.42	85.71±8.19	**97.79±6.33**	35.84±34.56	35.10±42.32	95.97±3.50	79.46±16.35	86.98±5.13	75.70±9.39	93.15±4.44
	13	69.56±11.74	72.59±15.11	**79.52±10.73**	7.58±12.74	18.81±22.16	76.64±12.74	62.99±13.49	48.79±16.02	45.26±12.23	62.98±9.24
	14	66.64±14.21	57.77±15.16	45.90±20.51	21.62±26.14	37.99±36.44	70.35±11.47	70.13±13.31	53.36±16.29	84.37±9.38	**87.42±7.32**
	15	84.57±10.12	65.38±9.01	39.20±10.36	33.87±36.54	28.39±33.88	79.95±4.57	76.63±14.97	72.76±15.99	90.25±7.72	**92.96±8.52**
	16	**92.88±6.69**	82.50±12.37	78.25±14.89	32.88±33.42	32.63±33.06	90.88±10.44	80.13±13.42	82.75±13.58	71.75±17.87	82.25±10.12
	17	61.82±14.72	54.55±19.43	60.00±15.64	9.70±17.24	12.42±26.85	73.03±19.30	42.73±22.70	49.70±21.68	87.88±11.69	**89.09±13.36**
	Unknown	40.74±38.59	16.79±22.41	36.93±26.65	36.88±40.50	59.77±37.01	63.91±36.12	40.62±33.30	32.29±28.93	28.19±36.44	**76.37±28.51**
	OA(%)	83.70±3.87	78.64±4.18	77.76±2.26	41.13±13.77	38.90±21.82	89.72±2.59	80.85±2.32	78.16±4.85	88.46±1.65	**92.98±1.07**
	AA(%)	79.06±3.76	71.27±2.48	68.16±2.98	29.94±8.44	33.59±18.43	83.41±5.00	70.47±3.83	68.98±3.66	87.49±1.37	**88.17±2.29**
	Kappa(*100)	80.98±4.32	75.16±4.55	74.00±3.27	35.32±13.29	33.80±21.77	87.25±3.90	77.50±2.68	74.51±5.35	86.27±1.97	**91.66±1.27**

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TABLE III
COMPARISON WITH OSR-BASED METHODS ON OTHER METRICS

	Methods	OSC-SCL	OSC_OCL	SSMLP-RPL	HyperTaFOR	DBEDML
NW	Training time(s)	17.10	89.937	480.77	565.52	505.83
	Testing time(s)	30.24	372.25	393.37	6.64	33.00
	FLOPs(G)	0.13	7.01	7.01	5.79	17.78
	Parameters(M)	1.89	6.43	6.44	0.26	50.78
TC	Training time(s)	26.92	48.72	430.38	560.42	1099.56
	Testing time(s)	51.78	866.36	410.96	18.32	92.32
	FLOPs(G)	0.13	7.01	7.01	5.79	17.78
	Parameters(M)	1.89	6.43	6.44	0.26	50.78
RP	Training time(s)	19.85	117.69	350.08	662.12	483.86
	Testing time(s)	33.52	777.77	280.98	13.70	45.25
	FLOPs(G)	0.13	7.01	7.01	7.24	20.74
	Parameters(M)	1.90	6.44	6.44	0.26	50.78
PU	Training time(s)	8.80	21.77	319.61	411.02	546.27
	Testing time(s)	6.77	175.17	230.15	2.64	29.76
	FLOPs(G)	0.11	5.14	6.68	4.88	5.84
	Parameters(M)	1.59	4.41	4.41	0.26	50.73
YC	Training time(s)	41.51	243.81	335.08	628.84	625.45
	Testing time(s)	2.86	55.44	58.55	88.65	6.85
	FLOPs(G)	0.27	22.08	22.08	12.49	31.27
	Parameters(M)	3.71	24.93	24.93	0.27	50.78

high accuracy on some land-cover classes. In self-constructed coastal wetland datasets, DBEDML excels in "Unknown" class recognition, ideal for mixed vegetation scenarios. For example, on NW, its "Unknown" accuracy is 77.81%, 1.85 times higher than HyperTaFOR and 5–9 times higher than OSC-SCL and SSFTT. On TC, DBEDML achieves 97.20% "Unknown" accuracy (the only model over 90%), 15.33% higher than the second-best HyperTaFOR with less fluctuation. CDFSL algorithms show poor accuracy and high variance, failing in complex environments. On RP, all models show high "Unknown" accuracy variability, but DBEDML is the only one above 80%, while others remain below. For public datasets, DBEDML attains 98.12% accuracy on PU with low fluctuation, far exceeding CSCANet. Other methods score below 50%, unable to handle urban unknowns. On YC, the "Unknown" accuracy of DBEDML is 16.60% higher than FDFSL and 12.46% higher than OSC-SCL, while others remain under 30%, indicating poor adaptation to wetland spectral variability and mixed ground objects.

Fig. 9 shows that DBEDML achieves optimal matching with the ground truth across all five datasets, with the white regions closely aligning with the ground truth.

Table III compares several OSR algorithms based on training time, testing time, FLOPs, and parameter count. OSC_OCL and SSMLP-RPL show moderate performance across all metrics, with small FLOPs and parameter counts within an acceptable range. OSC-SCL offers the best balance of efficiency and complexity, being lightweight across all metrics and ideal for scenarios with extremely limited resources. HyperTaFOR has the shortest testing time and is lightweight, but requires a lengthy training process. DBEDML lacks advantages in FLOPs and parameter count, resulting in longer training time. However, its testing time remains reasonable and faster than OSC_OCL and SSMLP-RPL, reflecting its performance.

E. Discussion

In this section, ablation experiments and optimal parameter determination for DBEDML are performed on the self-constructed NW, TC, and RP datasets.

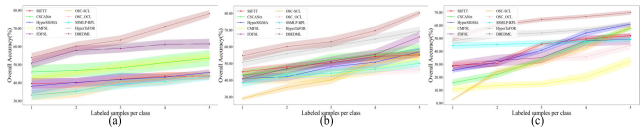


Fig. 10. OA (%) of different algorithms across three TD datasets with varying numbers of annotated samples. (a) NW. (b) TC. (c) RP.

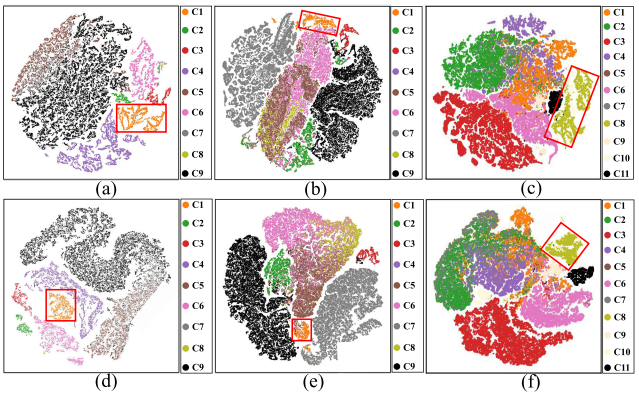


Fig. 11. t-SNE visualization of the original data and embedding features on the four datasets. (a) Original data of NW. (b) Original data of TC. (c) Original data of RP. (d) Embedding features of NW. (e) Embedding features of TC. (f) Embedding features of RP.

1) *Comparison of Different Numbers of Annotated Samples:* Fig. 10 shows OA of all algorithms with varying numbers of labeled samples across four datasets, with shaded bands indicating standard deviations. DBEDML consistently achieves the highest accuracy, significantly outperforming second-ranked methods like HyperTaFOR and FDFSL. Its moderate to low standard deviations demonstrate stable performance, resilience to randomness, and strong generalization. On the NW dataset, with five samples per class, DBEDML reaches 78.13%, surpassing HyperTaFOR (61.97%) and OSC-SCL (61.43%) by over 16 percentage points. With only one sample per class, DBEDML attains 53.96% accuracy with low variance. Similar trends appear on the TC and RP datasets. Notably, on the RP dataset, CMFSL and OSC-SCL fail, especially with one sample per class, while DBEDML exhibits extremely strong robustness.
2) *t-SNE Visualization of DBEDML:* Fig. 11 compares t-SNE visualizations before and after HVSE feature processing. HVSE reduces sample dispersion within categories, enhancing intraclass compactness, and decreases overlap between categories, improving interclass separation. The red-boxed representative points highlight these distribution changes. Dataset analysis shows NW and TC achieve strong interclass separation, while RP gains mainly in intraclass aggregation with limited interclass improvement. This limitation arises from easily confusable categories with minor spectral differences in RP, consistent with Fig. 9 and Table II(c), where RP attains the lowest classification accuracy.
3) *Comparison of Different λ:* λ in DBEDML controls the importance of different loss function components and ranges from 0.0 to 1.0 in increments of 0.05 or 0.1. Fig. 12 shows the OA, AA, and "Unknown" class

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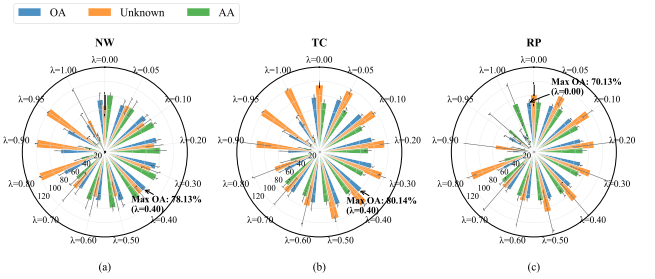


Fig. 12. Influence of lambda values on OA, Unknown(%), and AA across the three datasets. (a) NW. (b) TC. (c) RP.

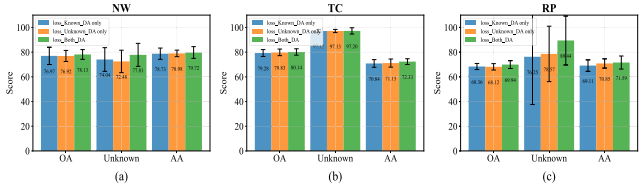


Fig. 13. Influence of $\mathcal{L}_{DA}^{\mathcal{K}}$ and $\mathcal{L}_{DA}^{\mathcal{U}}$ on OA, Unknown(%), and AA across the three datasets. (a) NW. (b) TC. (c) RP.

TABLE IV

ABLATION ANALYSIS OF THE COMPONENTS IN DBEDML ACROSS THREE DATASETS

Dataset	HVSE	Domain alignment Module	Energy-based Module	Embedding-wise Branch	Feature-wise Branch	OA(%)	Unknown(%)	AA(%)
NW	✔					58.15±8.76	59.81±24.25	64.49±5.33
	✔		✔	✔	✔	75.14±6.22	74.60±14.51	74.39±5.69
	✔	✔		✔	✔	74.42±6.52	75.08±17.71	73.85±9.56
	✔	✔	✔		✔	33.08±3.01	0.01±0.03	44.38±6.28
	✔	✔	✔	✔		73.36±6.70	64.82±14.69	81.75±1.83
	✔	✔	✔	✔	✔	**78.13±3.94**	**77.81±11.31**	**79.72±4.75**
TC	✔					51.87±7.82	81.75±1.83	60.82±7.69
	✔		✔	✔	✔	76.65±2.67	94.06±4.03	69.22±2.23
	✔	✔		✔	✔	77.59±2.73	95.94±5.70	69.83±3.14
	✔	✔	✔		✔	45.27±4.63	6.81±18.01	43.69±6.86
	✔	✔	✔	✔		74.02±2.60	90.45±4.36	62.83±3.22
	✔	✔	✔	✔	✔	**80.14±2.59**	**97.20±2.56**	**72.33±2.33**
RP	✔					64.97±3.91	54.17±32.95	60.80±9.34
	✔		✔	✔	✔	65.89±2.44	84.38±29.80	67.92±4.33
	✔	✔		✔	✔	66.83±2.51	86.91±29.66	69.60±4.76
	✔	✔	✔		✔	60.51±6.19	11.23±27.32	45.56±4.94
	✔	✔	✔	✔		68.61±2.17	65.80±40.66	70.81±3.19
	✔	✔	✔	✔	✔	**69.94±3.17**	**89.44±29.82**	**71.59±5.29**

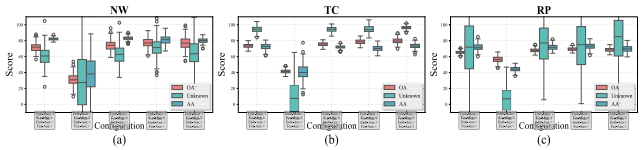


Fig. 14. Influence of main classification task and auxiliary contrastive learning task in dual branch on OA, Unknown(%), and AA across the three datasets. (a) NW. (b) TC. (c) RP.

accuracy across three datasets. For NW, all metrics rise steadily, peak at $\lambda = 0.4$, then sharply decline, with high volatility beyond 0.5. The TC dataset follows a similar pattern, peaking at 0.4 but with smaller fluctuations and more stability at higher λ values. RP, characterized by high class homogeneity, shows no clear peak and poor stability, achieving its highest OA at $\lambda = 0.0$. The "Unknown" class accuracy in RP fluctuates wildly, e.g., 96.54% at 0.8 but dropping to 29.51% at 0.95, reflecting ambiguous boundaries between known and unknown classes. To prevent the model from overfitting, λ is set to 0.4.

4) *Ablation Study of $\mathcal{L}_{DA}^{\mathcal{K}}$ and $\mathcal{L}_{DA}^{\mathcal{U}}$:* $\mathcal{L}_{DA}^{\mathcal{K}}$ and $\mathcal{L}_{DA}^{\mathcal{U}}$ represent losses for known-class and unknown-class feature alignment in the DA process, together forming $\mathcal{L}_{DA}$. Fig. 13 shows that using $\mathcal{L}_{DA}$ yields higher OA than using either component alone, with the smallest gain on TC and the largest on RP, indicating RP demands a more comprehensive DA strategy. For the "Unknown" metric on NW, $\mathcal{L}_{DA}$ improves by 3.77% over $\mathcal{L}_{DA}^{\mathcal{K}}$ and 5.33% over $\mathcal{L}_{DA}^{\mathcal{U}}$. Improvements on TC are minimal, while on RP, $\mathcal{L}_{DA}$ boosts performance by 13.19% and 10.87%, respectively. Across datasets, AA varies little, showing it depends more on basic capability than DA strategy. In addition, $\mathcal{L}_{DA}^{\mathcal{U}}$ outperforms $\mathcal{L}_{DA}^{\mathcal{K}}$, excelling in the "Unknown" metric and offering better stability, suggesting that in CDFSOSR tasks, addressing the "Unknown" class distribution is more critical than handling the known class distribution.

5) *Ablation Study of DBEDML:* Taking HVSE as the benchmark, ablation experiments are conducted. HVSE adopts a hard-threshold mechanism for OSR, and the corresponding experimental results are presented in Table IV. The results obtained only on HVSE are unsatisfactory, with substantial declines observed for all three metrics across the three datasets. The EWB is the most critical, as its removal causes severe performance drops, especially in "Unknown" class recognition, where accuracy nearly vanishes in NW and TC datasets. This performance drop arises as the EWB, the core module of DBEDML that underpins its OSR capability, learns compact discriminative manifolds for known classes. Ablating the EWB causes severe overlap between known and novel class distributions, disabling the EBM from reliable known and unknown discrimination. The DA Module consistently improves OA by 2.99%–4.05%, highlighting cross-domain consistency as essential. The FWB significantly boosts "Unknown" accuracy by 23.64% in the challenging RP dataset. The EBM provides stable improvements across all metrics. The full model achieves optimal performance in all nine evaluation scenarios, showing synergistic effects beyond simple addition. These results indicate that effective CDFSOSR requires balanced integration of embedding, feature refinement, energy optimization, and DA, with component importance varying by dataset.

6) *Ablation Study of the Two Tasks in Dual Branch:* Both EWB and FWB include two tasks, namely the main classification task and the auxiliary contrastive learning task. Ablation experiments verify the roles of these four tasks, with results shown in Fig. 14. The boxplots reveal that Config 1 significantly outperforms Config 2 across all datasets and metrics, indicating Emb-Main is more effective than Fea-Main. Config 2 performs poorly on the "Unknown" metric and shows low stability, suggesting Fea-Main struggles with "Unknown" classes when used alone. Config 3 slightly improves over Config 1 and substantially over Config 2, demonstrating that combining Emb-Main and Fea-Main is more effective than either alone. Adding the auxiliary contrastive learning task, Configs 4 and 5 further improve OA and "Unknown" metrics compared to Config 3, though with a slight decrease in AA.

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Fea-Aux excels in recognizing "Unknown" classes, while Emb-Aux offers greater OA stability.

V. Conclusion

This article, for the first time, introduces DBEDML, a dual-branch energy-driven meta-learning framework for the CDFSOSR task. To tackle the open-set challenge in the FSL, we employ contrastive learning and distance metrics to combine real prototypes with known class features while repelling fake prototypes. An energy function is designed to improve discrimination between known and unknown class features. To solve the DA difficulties caused by label space inconsistencies between SD and TD, we incorporate fake prototypes of SD as unknown classes, enabling effective alignment of known domains and rejection of unknown classes in TD. Extensive experiments on three self-constructed coastal wetland datasets and two public datasets to verify the effectiveness of the proposed DBEDML.

In future research, more attention should be paid to the impact of the open-set problem on domain generalization. Specifically, the dual-branch energy-driven meta-learning paradigm is leveraged to learn domain-invariant energy metrics across multiple SDs during training, which enables the well-trained model to directly generalize to completely unseen TDs without any TD data. This task will be more in line with real-world scenes, yet it will also face more challenges.

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Qixing Yu received the master's degree in software engineering from the College of Computer Science and Technology, Qingdao University, Qingdao, China, in 2024. He is currently pursuing the Ph.D. degree in marine resources and information engineering with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao.

His recent research focuses on the intelligent processing of hyperspectral remote sensing images.



Zhongwei Li received the Ph.D. degree from China University of Petroleum (East China), Qingdao, China, in 2011.

He is currently a Professor with the College of Oceanography and Space Informatics, China University of Petroleum (East China). His research interests include remote sensing image processing, ocean numerical forecasting, and cloud computing.



Fangming Guo (Member, IEEE) received the Ph.D. degree from China University of Petroleum (East China), Qingdao, China, in 2024.

He is currently a Post-Doctoral Researcher with the College of Oceanography and Space Informatics, China University of Petroleum (East China). His research interests include the intelligent processing of remote sensing information and wetland remote sensing monitoring.



Ziqi Xin received the B.E. degree in communication engineering from China University of Petroleum (East China), Qingdao, China, in 2020, where he is currently pursuing the Ph.D. degree in computer technology and resource information engineering with the College of Oceanography and Space Informatics.

His research focuses on hyperspectral image classification.



Guangbo Ren received the Ph.D. degree in marine information detection and processing from the Ocean University of China, Qingdao, China, in 2010.

He is currently an Associate Professor with the First Institute of Oceanography, Ministry of Natural Resources, Qingdao. His research interests include ocean ecosystem hyperspectral remote sensing and coastal dynamics remote sensing.



Jianbu Wang received the master's degree from China University of Petroleum (East China), Qingdao, China, in 2008.

He is currently working toward the Ph.D. degree in marine resources and information engineering with the College of Oceanography and Space Informatics, China University of Petroleum (East China). He is mainly engaged in research on remote sensing monitoring methods for coastal wetlands.



Zhenggang Bi received the B.S. degree in landscape architecture from Qingdao Agricultural University, Qingdao, China, in 2018.

He is currently an Engineer with Shandong Yellow River Delta National Nature Reserve, Dongying. His research interests include wetland ecosystem monitoring, remote sensing applications in ecological conservation, and biodiversity assessment in coastal wetland areas.

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